METRIC

MIL-M-28787/459

3 December 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ENCODER, PARITY, DIGITAL, MODULE, KEY CODE JPJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic parity encoder module.

1.2 Classification. This specification covers one class of module. Class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.21009 failures per million operating hours for class I when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JPJ.

3.6.2 Size. The module size shall be 1A Format A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 40.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	INF	10	INF	10	INF	MOhms
B	Vcc stress	IV	Q	---------Module shall pass all remaining tests----------						
C1-C33	IBD	V	S	-0.01	0.1	-0.01	0.1	-0.01	0.1	mA
D1-D5	Output short ckt current	VI	S	-15	-100	-15	-100	-15	-100	mA
E1-E19	Functional	VII	X	---Test vectors must produce output vectors as --- -----specified in table VIII.-----						
F1-F33	Input Clamp Diode Volt.	VIII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G5	Output Volt. (low level)	IX	X	0	400	0	400	0	400	mV
H1-H5	Output Volt. (high level)	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J33	Input Cur. (high level)	XI	X	-2	20	-2	20	-2	20	uA
K1-K33	Input Cur. (low level)	XII	X	-0.13	-0.37	-0.13	-0.37	-0.13	-0.37	mA
L	Power Supply Current	XIII	X	0	100	0	100	0	100	mA
N1-N21	Propagation Delay Time (tPLH) CL=50pF	XIV	S	2	55	2	55	2	55	ns

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
M22-M27	Propagation Delay Time (tPLH) CL=50pF	XIV	S	2	110	2	110	2	110	nS
M28-M48	Propagation Delay Time (tPLH) CL=150pF	XIV	Q	2	65	2	65	2	65	nS
M49-M54	Propagation Delay Time (tPLH) CL=150pF	XIV	Q	2	130	2	130	2	130	nS
N1-N23	Propagation Delay Time (tPHL) CL=50pF	XV	S	2	50	2	50	2	50	nS
N24-N29	Propagation Delay Time (tPHL) CL=50pF	XV	S	2	100	2	100	2	100	nS
N30-N52	Propagation Delay Time (tPHL) CL=150pF	XV	Q	2	60	2	60	2	60	nS
N53-N58	Propagation Delay Time (tPHL) CL=150pF	XV	Q	2	120	2	120	2	120	nS

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
P1-P3	Output xtalk (low level)	XVI	Q	0	500	0	500	0	500	mV
Q1-Q3	Output xtalk (high level)	XVI	Q	0	400	0	400	0	400	mV
R	PSTR	XVII	Q	0	400	0	400	0	400	mV
S	PSFC	XVIII	S	1.6	3.0	1.6	3.0	1.6	3.0	uF
T1-T33	Input Cap.	XIX	Q	2	15	2	15	2	15	pF

-See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/459-1 for class I, except no marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

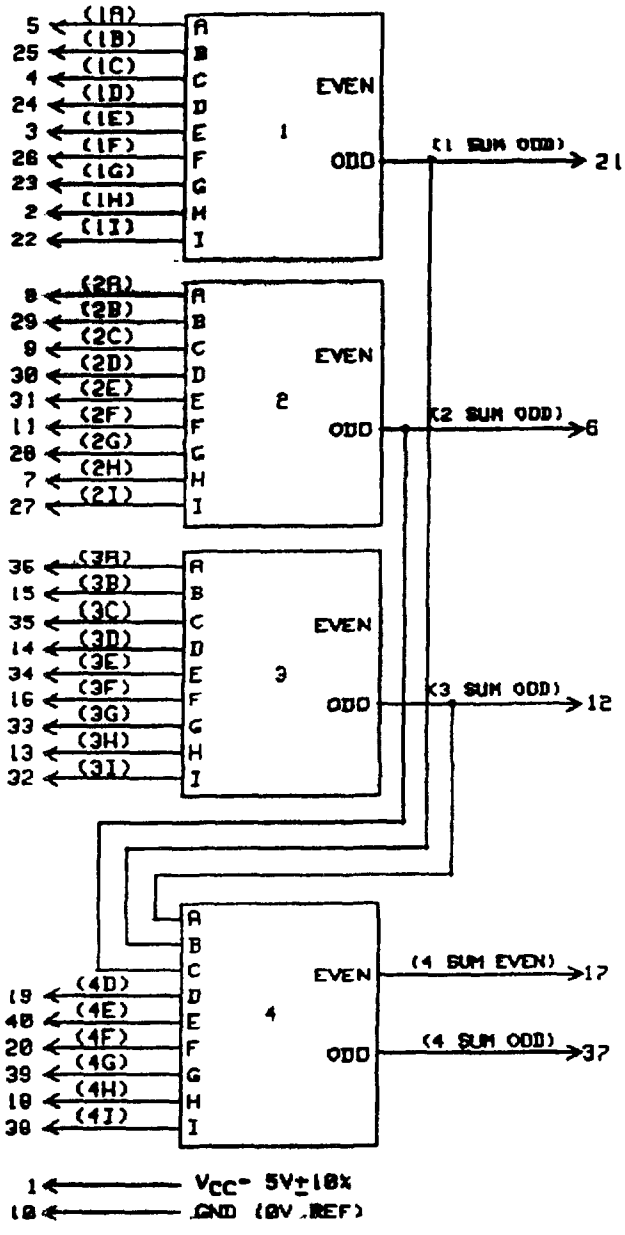
3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 4.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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NOTES:
1. 1A-1I, 2A-2I, 3A-3I, 4D-4I = Data inputs.
2. 1 sum odd, 2 sum odd, 3 sum odd, 4 sum odd = Odd parity outputs.
3. 4 sum even = Even parity output.

FIGURE 1. JPJ functional block diagram.

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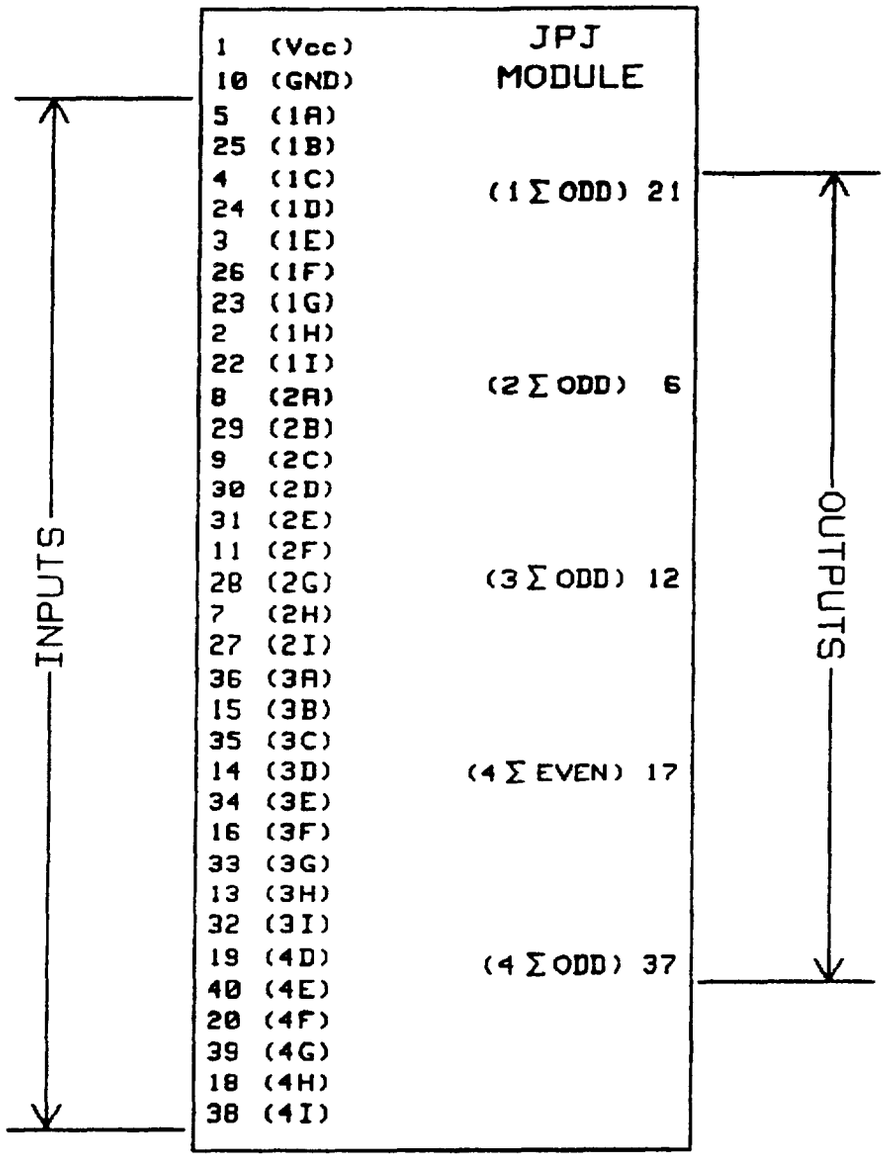


FIGURE 2. JPJ pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and F are stress tests and shall be performed only at 25 $\pm$ 5°C. If test conditions A, B, C, D, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, and F where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E and G through S shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25 ± 5°C ambient.

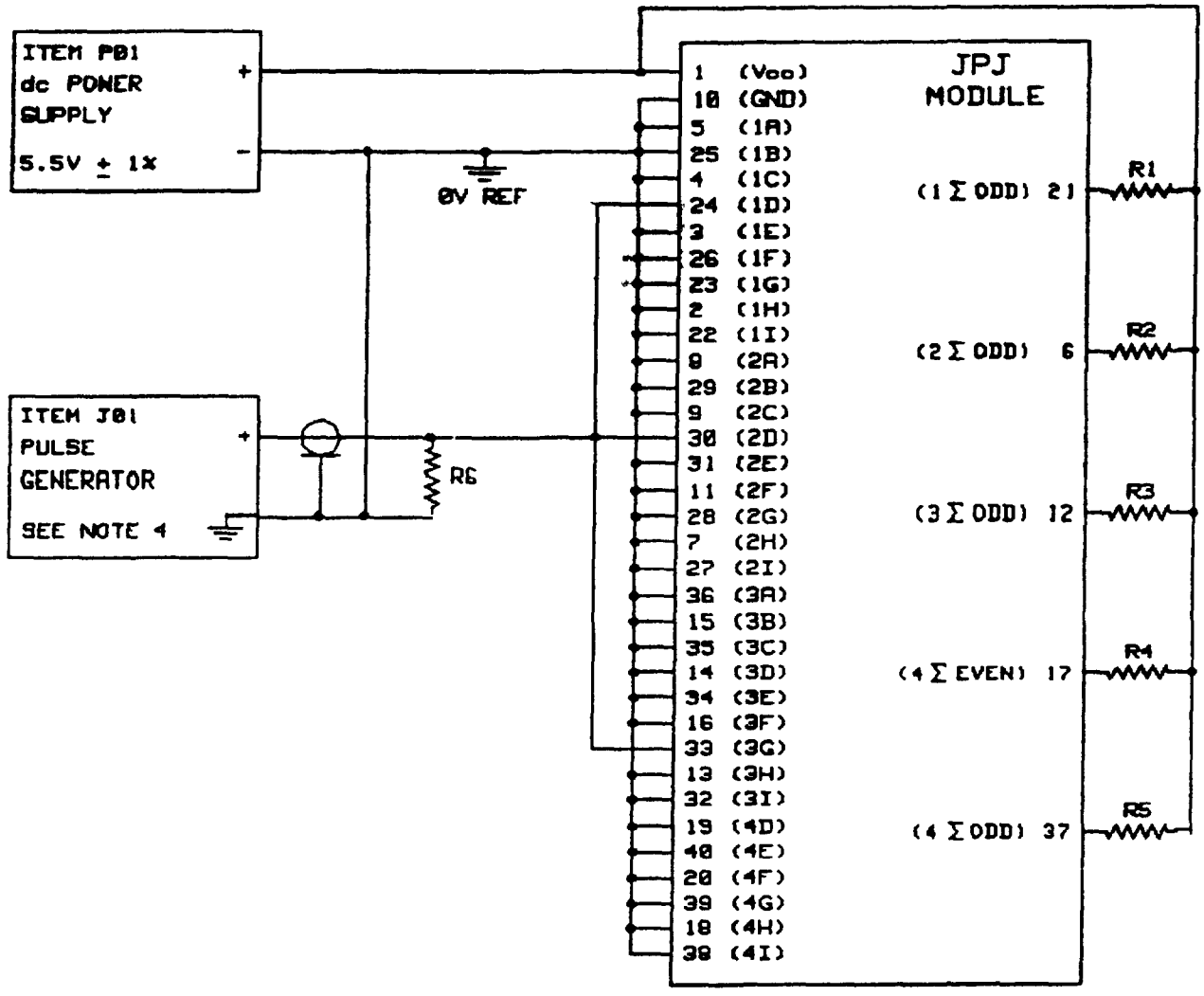
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests E and G through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests E and G through S shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

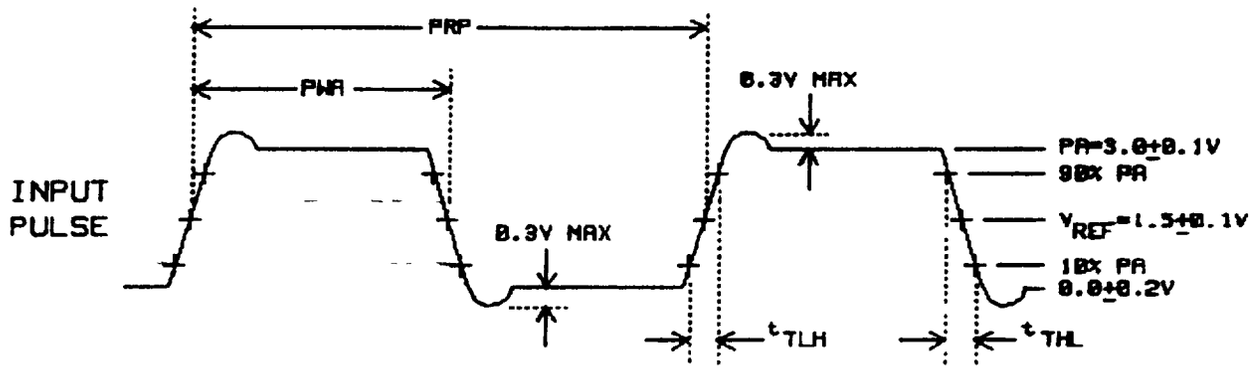
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NOTES:
1. R1 = R2 = R3 = 385 Ohms ± 1 percent, 1/4W.
2. R4 = R5 = 350 Ohms ± 1 percent, 1/4W.
3. R6 = 51.1 Ohms ± 1 percent, 1/4W.
4. See figure 4 for operating temperature and life test waveform.

FIGURE 3. JPJ operating temperature and life test circuit.

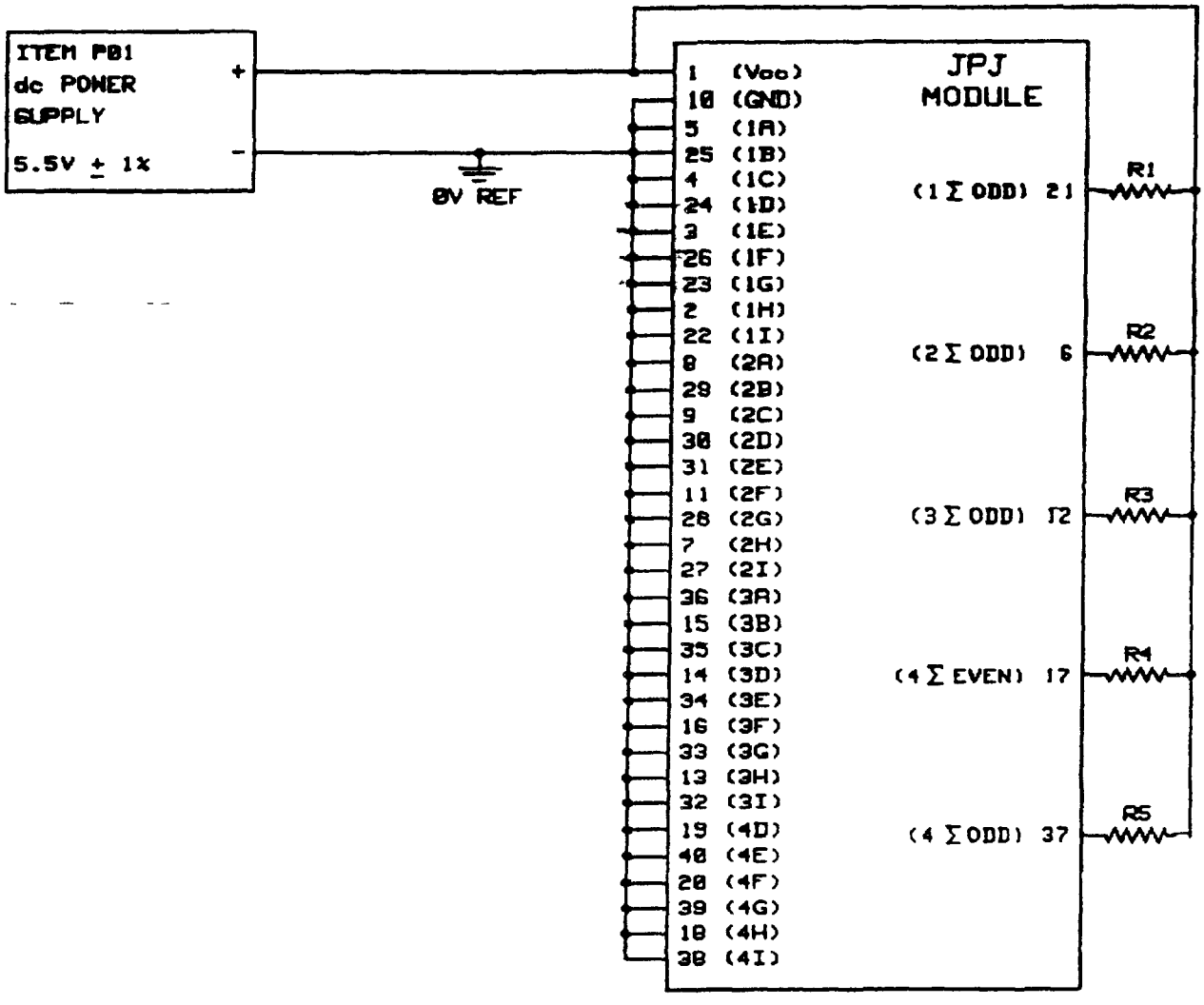
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NOTE:
Input pulse parameters:
tTLH = tTHL ≤ 6 ± 2 ns.
PRP = 1000 ± 200 ns.
PWA = 500 ± 100 ns.

FIGURE 4. JPJ operating temperature and life test waveform.

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NOTES:
1. R1 = R2 = R3 = 385 Ohms ± 1 percent, 1/4W.
2. R4 = R5 = 350 Ohms ± 1 percent, 1/4W.

FIGURE 5. JPJ component temperature test circuit.

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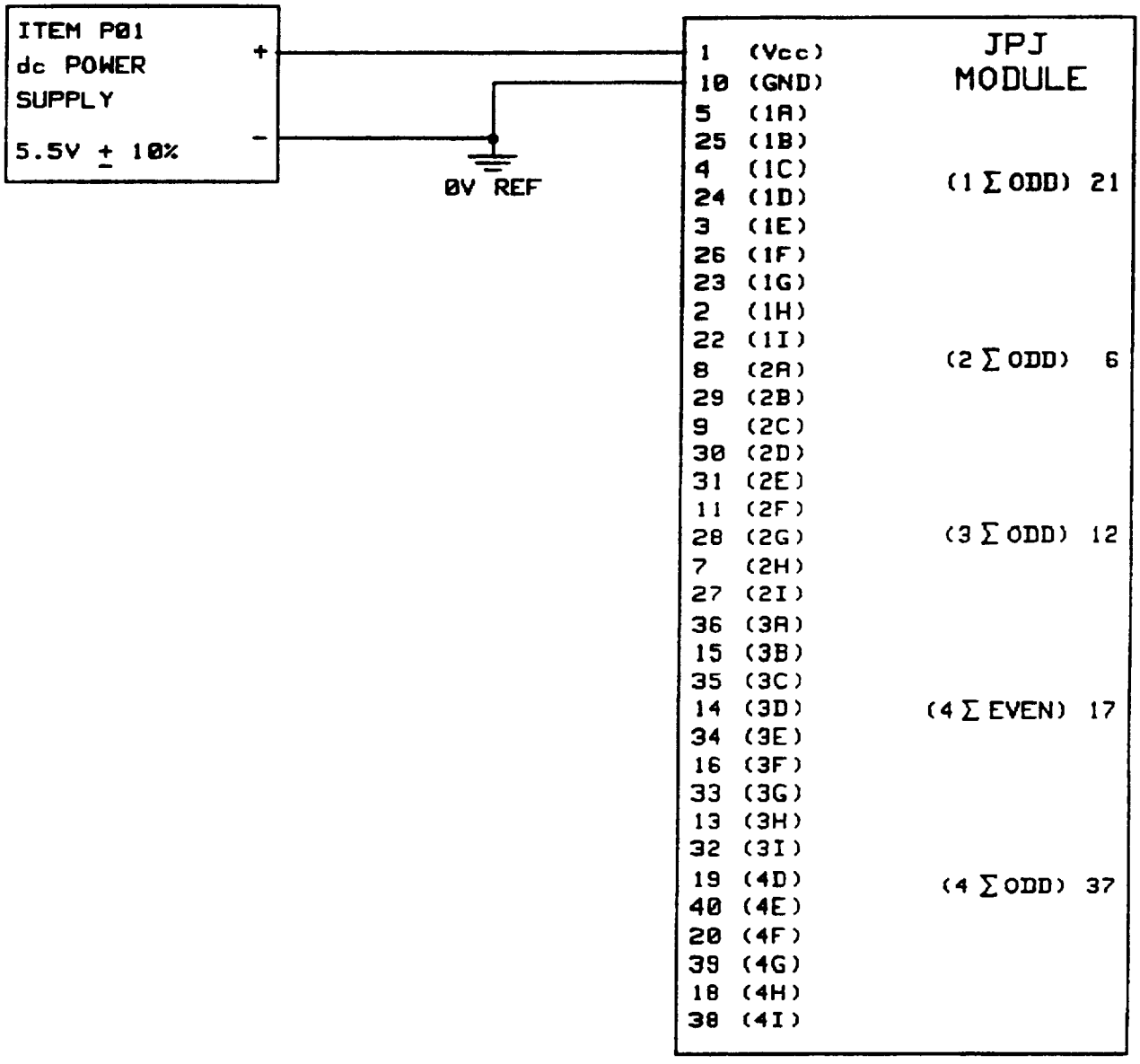


FIGURE 6. JPJ vibration test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Ioff = open drain leakage current.
(i) Ios = output short circuit current.
(j) Iozh = output high level leakage current.
(k) Iozl = output low level leakage current.
(l) PSFC = power supply filter capacitance.
(m) PSTA = power supply transient amplitude.
(n) Vic = input clamp diode voltage.
(o) Voh = output high level voltage.
(p) Vol = output low level voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V to 5.5 Vdc. Module power dissipation: 594 mW (max).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
1A,1B,1C,1D, 1E,1F,1G,1H,1I	-0.4	0.7	2.0	-0.12	0.02	0.1	15
2A,2B,2C,2D, 2E,2F,2G,2H,2I	-0.4	0.7	2.0	-0.12	0.02	0.1	15
3A,3B,3C,3D, 3E,3F,3G,3H,3I	-0.4	0.7	2.0	-0.12	0.02	0.1	15
4D,4E,4F,4G, 4H,4I	-0.4	0.7	2.0	-0.12	0.02	0.1	15

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
4 SUM EVEN, 4 SUM ODD	2.5	0.4	-0.4	4.0	-100	NA
1 SUM ODD, 2 SUM ODD, 3 SUM ODD	2.5	0.4	-0.38	3.63	-100	NA

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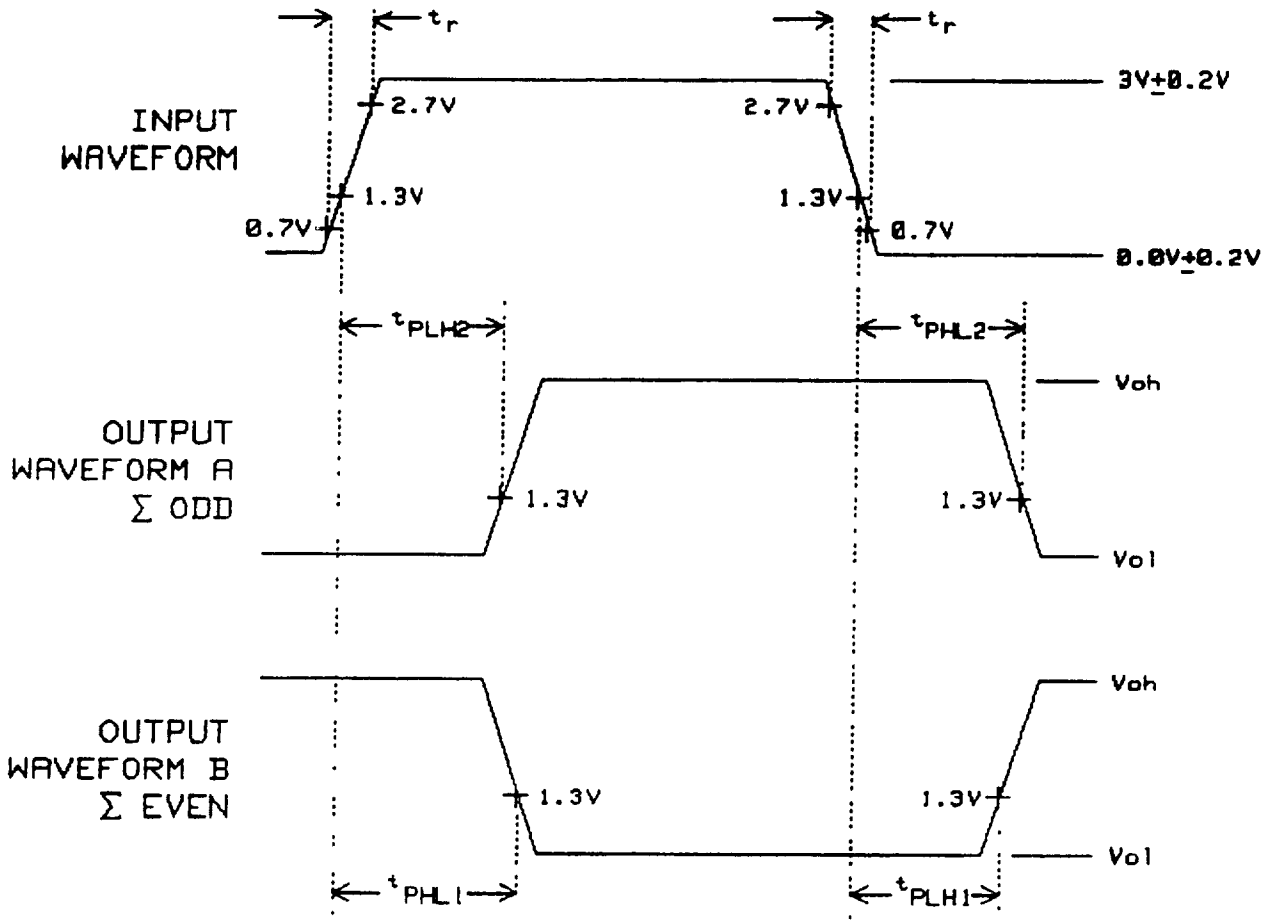


FIGURE 6a. Dynamic waveforms.

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6.4 Module function. This module contains four 9-bit parity encoders interconnected as one 33-bit parity encoder.

6.4.1 Input to output relationship. Each parity encoder has the input to output relationship in table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/

Number of inputs A thru I** that are high	Outputs	
	Sum even	Sum odd
0, 2, 4, 6, 8	H	L
1, 3, 5, 7, 9	L	H

1/ H = High-voltage level.
2/ L = Low-voltage level.
3/ ** = Parity encoder 4 obtains inputs A, B, C from 3 sum odd, 1 sum odd and 2 sum odd, respectively.

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0079)

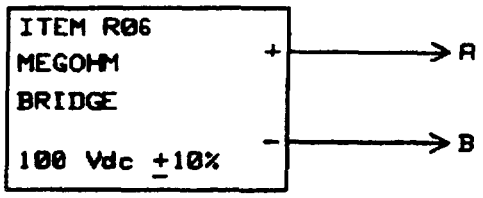
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TABLE III. Isolation test.

Test cond	Test description	Set-up conditions		25 ± 5°C Initial limits		Units	Test fig
		Connect A to	Connect B to	Min	Max		
A	Isolation	All pins	FRAME	10	INF	MOhms	7

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JPJ MODULE		
1	(Vcc)	
10	(GND)	
5	(1A)	
25	(1B)	
4	(1C)	
24	(1D)	(1 Σ ODD) 21
3	(1E)	
26	(1F)	
23	(1G)	
2	(1H)	
22	(1I)	
8	(2A)	(2 Σ ODD) 6
29	(2B)	
9	(2C)	
30	(2D)	
31	(2E)	
11	(2F)	
28	(2G)	(3 Σ ODD) 12
7	(2H)	
27	(2I)	
36	(3A)	
15	(3B)	
35	(3C)	
14	(3D)	(4 Σ EVEN) 17
34	(3E)	
16	(3F)	
33	(3G)	
13	(3H)	
32	(3I)	
19	(4D)	(4 Σ ODD) 37
40	(4E)	
20	(4F)	
39	(4G)	
18	(4H)	
38	(4I)	

NOTES:

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

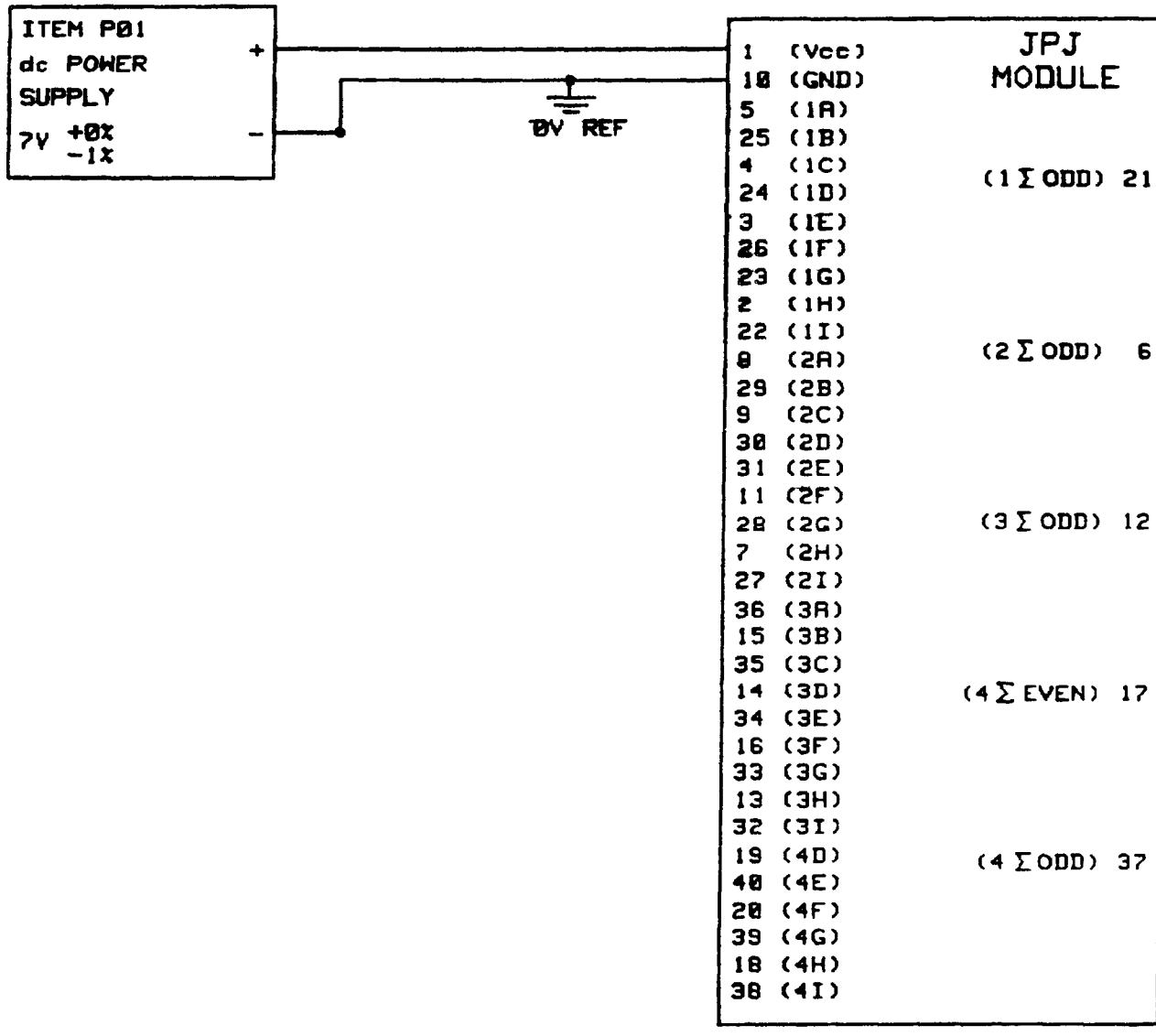
FIGURE 7. Isolation test circuit for table III.

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TABLE IV. Vcc stress test.

Test cond	Signal name	25 ± 5°C Initial limits Min	Max	Test fig
B	Vcc stress	Module shall pass all remaining tests.		8

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NOTE:

1. The duration of this test shall be maintained for a minimum of 5 seconds.

FIGURE 8. Vcc stress.

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TABLE V. Input breakdown current test.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
C1	1A	2,3,4,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	5	-0.01	0.1	mA	9
C2	1B	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	25				
C3	1C	2,3,5,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	4				
C4	1D	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	24				
C5	1E	2,4,5,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	3				
C6	1F	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,27,28,29, 30,31,32,33,34, 35,36,38,39,40	26	-0.01	0.1	mA	9

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
C7	1G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	23	-0.01	0.1	mA	9
C8	1H	3,4,5,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	2				
C9	1I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	22				
C10	2A	2,3,4,5,7,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	8				
C11	2B	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 30,31,32,33,34, 35,36,38,39,40	29				
C12	2C	2,3,4,5,7,8,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	9	-0.01	0.1	mA	9

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
C13	2D	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,31,32,33,34, 35,36,38,39,40	30	-0.01	0.1	mA	9
C14	2E	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,32,33,34, 35,36,38,39,40	31				
C15	2F	2,3,4,5,7,8,9, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	11				
C16	2G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,29, 30,31,32,33,34, 35,36,38,39,40	28				
C17	2H	2,3,4,5,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	7				
C18	2I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,28,29, 30,31,32,33,34, 35,36,38,39,40	27	-0.01	0.1	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:18Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
C19	3A	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,38,39,40	36	-0.01	0.1	mA	9
C20	3B	2,3,4,5,7,8,9, 11,13,14,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	15				
C21	3C	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,36,38,39,40	35				
C22	3D	2,3,4,5,7,8,9, 11,13,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	14				
C23	3E	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 35,36,38,39,40	34				
C24	3F	2,3,4,5,7,8,9, 11,13,14,15,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	16	-0.01	0.1	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:18Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

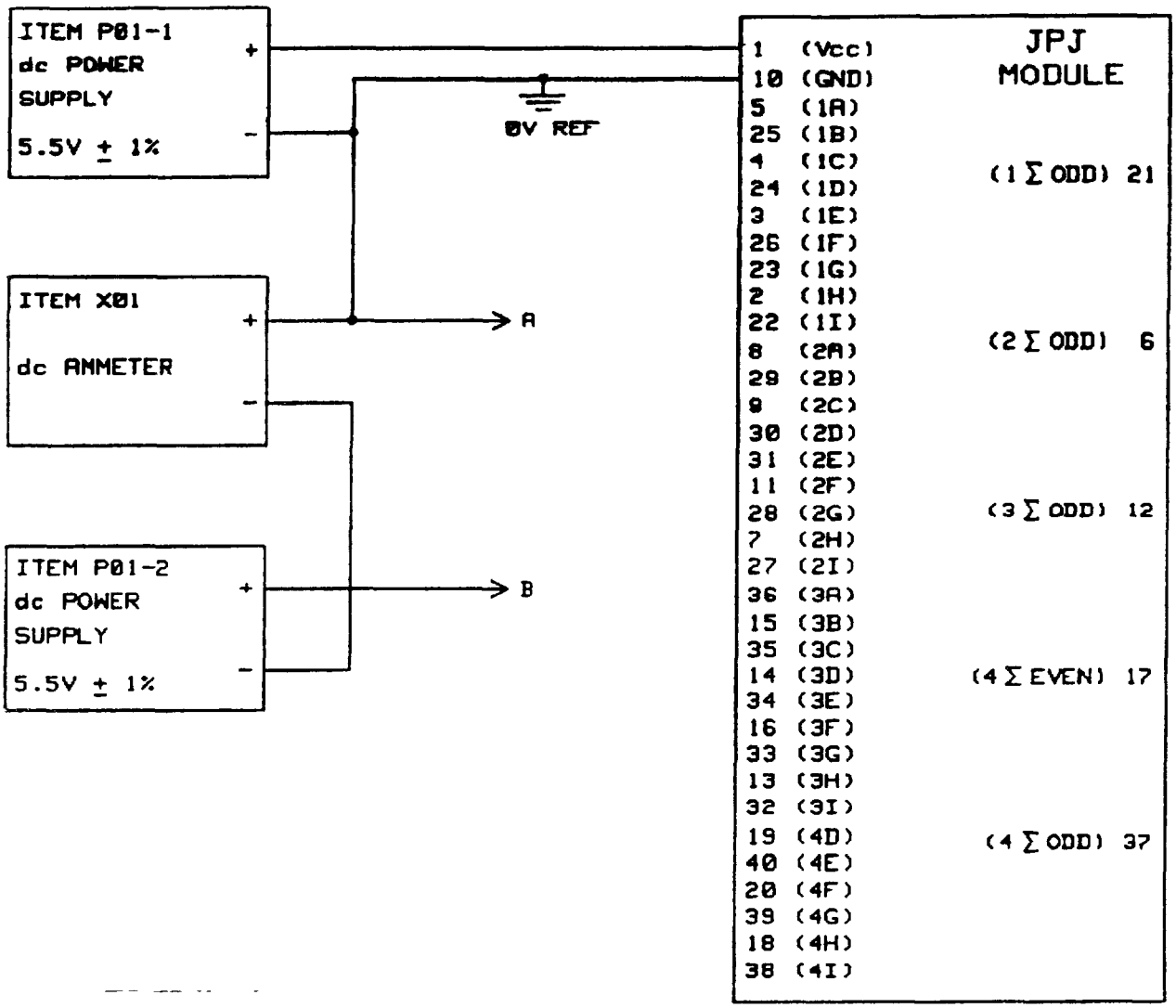
Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
C25	3G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,34, 35,36,38,39,40	33	-0.01	0.1	mA	9
C26	3H	2,3,4,5,7,8,9, 11,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	13				
C27	3I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,33,34, 35,36,38,39,40	32				
C28	4D	2,3,4,5,7,8,9, 11,13,14,15,16, 18,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	19				
C29	4E	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,36,38,39	40				
C30	4F	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	20	-0.01	0.1	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:18Z
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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	Max	Units	Test fig
C31	4G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,36,38,40	39	-0.01	0.1	mA	9
C32	4H	2,3,4,5,7,8,9, 11,13,14,15,16, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	18				
C33	4I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,36,39,40	38	-0.01	0.1	mA	9

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:18Z
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NOTES:

1. Caution: Do not isolate the chassis of item X01 from earth ground.

2. Item P01-2 shall be current limited to 5mA.

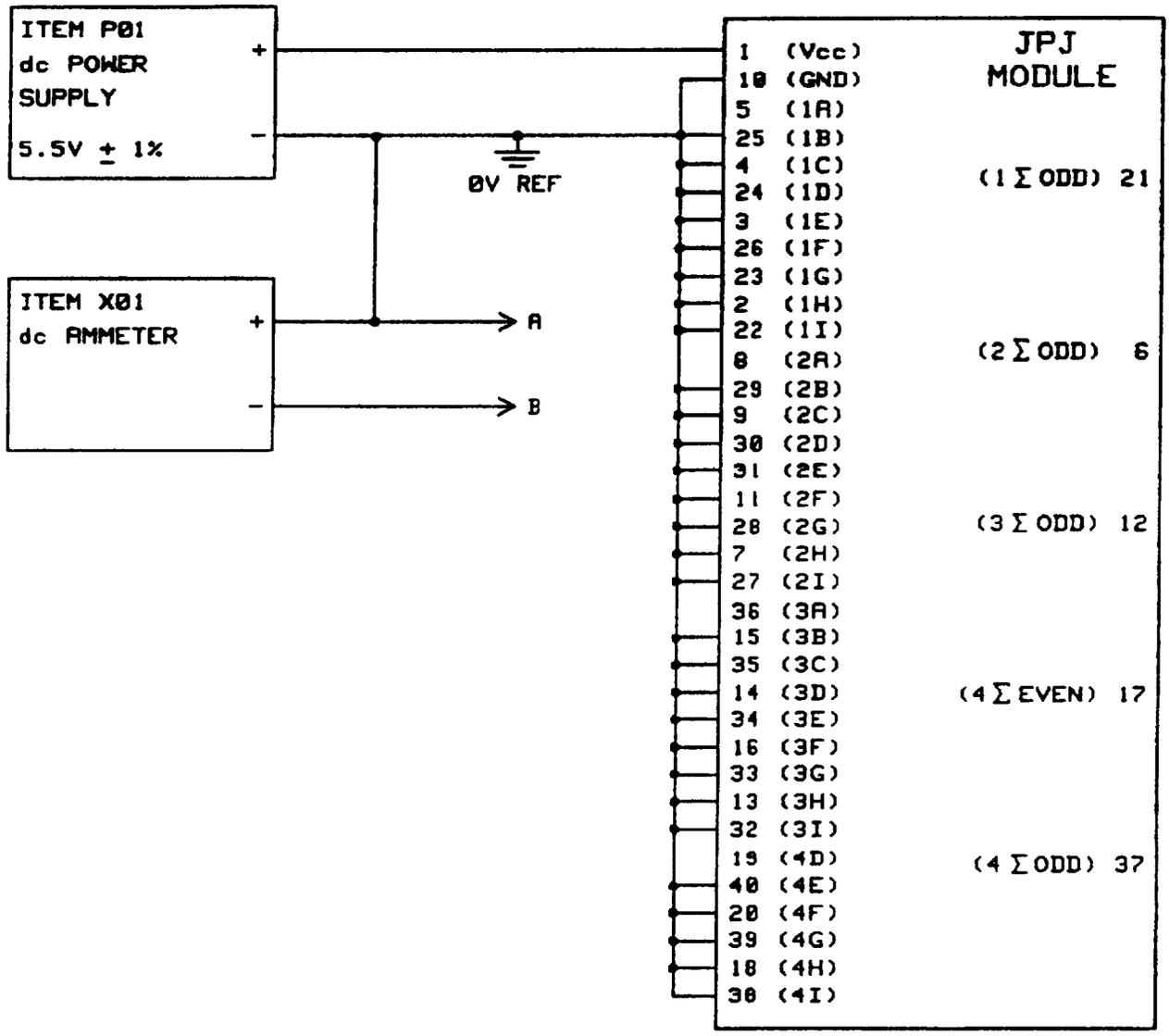
FIGURE 9. Input breakdown test circuit for table V.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
D1	1 SUM ODD	8,19,36	21	-15	-100	mA	10
D2	2 SUM ODD	5,19,36	6				
D3	3 SUM ODD	5,8,19	12				
D4	4 SUM ODD	5,8,36	37				
D5	4 SUM EVEN	5,8,19,36	17	-15	-100	mA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:18Z
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NOTES:

1. Caution: Do not isolate the chassis of item X01 from earth ground.

2. When performing output short circuit current tests, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

FIGURE 10. Output short circuit test circuit for table VI.

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TABLE VII. Functional test.

Test cond	A to pin	B to pin	C to pin	1/ Output logic states 1 sum odd	2 sum odd	3 sum odd	4 sum odd	4 sum even	Test fig
E1	4,8,11,13, 15,16,23, 24,25,29, 30,32,34, 36	2,3,5,7,9, 14,18,19, 20,22,26, 27,28,31, 33,35,38, 39,40	2/3/	L	L	L	L	H	11
E2	2,3,8,9,15, 18,19,22, 24,28,30, 31,32,33, 35,39	4,5,7,11, 13,14,16, 20,23,25, 26,27,29, 34,36,38,40		L	H	L	L	H	
E3	2,5,11,13, 15,22,24, 25,27,28, 30,31,32, 34,38,39	3,4,7,8,9, 14,16,18, 19,20,23, 26,29,33, 35,36,40		H	H	L	L	H	
E4	4,5,7,8,11, 14,15,16, 18,22,23, 24,25,26, 27,30,32, 33,34	2,3,9,13, 19,20,28, 29,31,35, 36,38,39, 40		H	H	L	H	L	
E5	3,4,5,9,11, 14,15,19, 23,24,27, 28,30,31, 35	2,7,8,13, 16,18,20, 22,25,26, 29,32,33, 34,36,38, 39,40		H	L	H	H	L	
E6	5,7,8,9,11, 13,14,15, 16,18,20, 22,24,25, 27,31,34, 35,38,39,40	2,3,4,19, 23,26,28, 29,30,32, 33,36	2/3/	L	L	L	H	L	11

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:18Z
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TABLE VII. Functional test - Continued.

Test cond	A to pin	B to pin	C to pin	1/ Output logic states 1 sum odd	2 sum odd	3 sum odd	4 sum odd	4 sum even	Test fig
E7	3,5,7,8,9, 13,14,15, 20,25,26, 27,28,29, 33,35,38	2,4,11,16, 18,19,22, 23,24,30, 31,32,34, 36,39,40	2/3/	L	L	H	H	L	11
E8	2,3,5,7,9, 13,15,16, 18,23,24, 25,26,27, 28,29,30, 34,35,36,38	4,8,11,14, 19,20,22, 31,32,33, 39,40		H	L	L	H	L	
E9	3,5,8,11, 13,14,18, 20,23,24, 29,30,35, 36,38,39	2,4,7,9,15, 16,19,22, 25,26,27, 28,31,32, 33,34,40		L	L	L	L	H	
E10	2,8,9,11, 16,18,19, 20,22,23, 24,25,26, 28,32,38, 40	3,4,5,7,13, 14,15,27, 29,30,31, 33,34,35, 36,39		L	L	L	H	L	
E11	2,8,9,11, 15,16,22, 23,25,27, 29,30,33, 34,35,39	3,4,5,7,13, 14,18,19, 20,24,26, 28,31,32, 36,38,40		L	L	H	L	H	
E12	4,5,9,13, 14,15,16, 19,20,26, 27,28,31, 35,38	2,3,7,8,11, 18,22,23, 24,25,29, 30,32,33, 34,36,39,40	2/3/	H	L	H	H	L	11

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:18Z
Check the source to verify that this is the current version before use.

TABLE VII. Functional test - Continued.

Test cond	A to pin	B to pin	C to pin	1/ Output logic states 1 sum odd	2 sum odd	3 sum odd	4 sum odd	4 sum even	Test fig
E13	4,5,7,8,13, 14,18,19, 24,32,33, 34,36,38,39	2,3,9,11, 15,16,20, 22,23,25, 26,27,28, 29,30,31, 35,40	2/3/	H	L	L	H	L	11
E14	4,5,7,8,9, 14,15,16, 20,22,28, 30,33,39	2,3,11,13, 18,19,23, 24,25,26, 27,29,31, 32,34,35, 36,38,40		H	H	L	L	H	
E15	2,3,11,13, 14,16,18, 20,23,25, 27,28,29, 30,31,32, 36,38	4,5,7,8,9, 15,19,22, 24,26,33, 34,35,39,40		L	L	H	L	H	
E16	4,5,7,16, 18,22,23, 24,25,29, 33,35	2,3,8,9,11, 13,14,15, 19,20,26, 27,28,30, 31,32,34, 36,38,39,40	2/3/	L	L	H	L	H	11

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:18Z
Check the source to verify that this is the current version before use.

TABLE VII. Functional test - Continued.

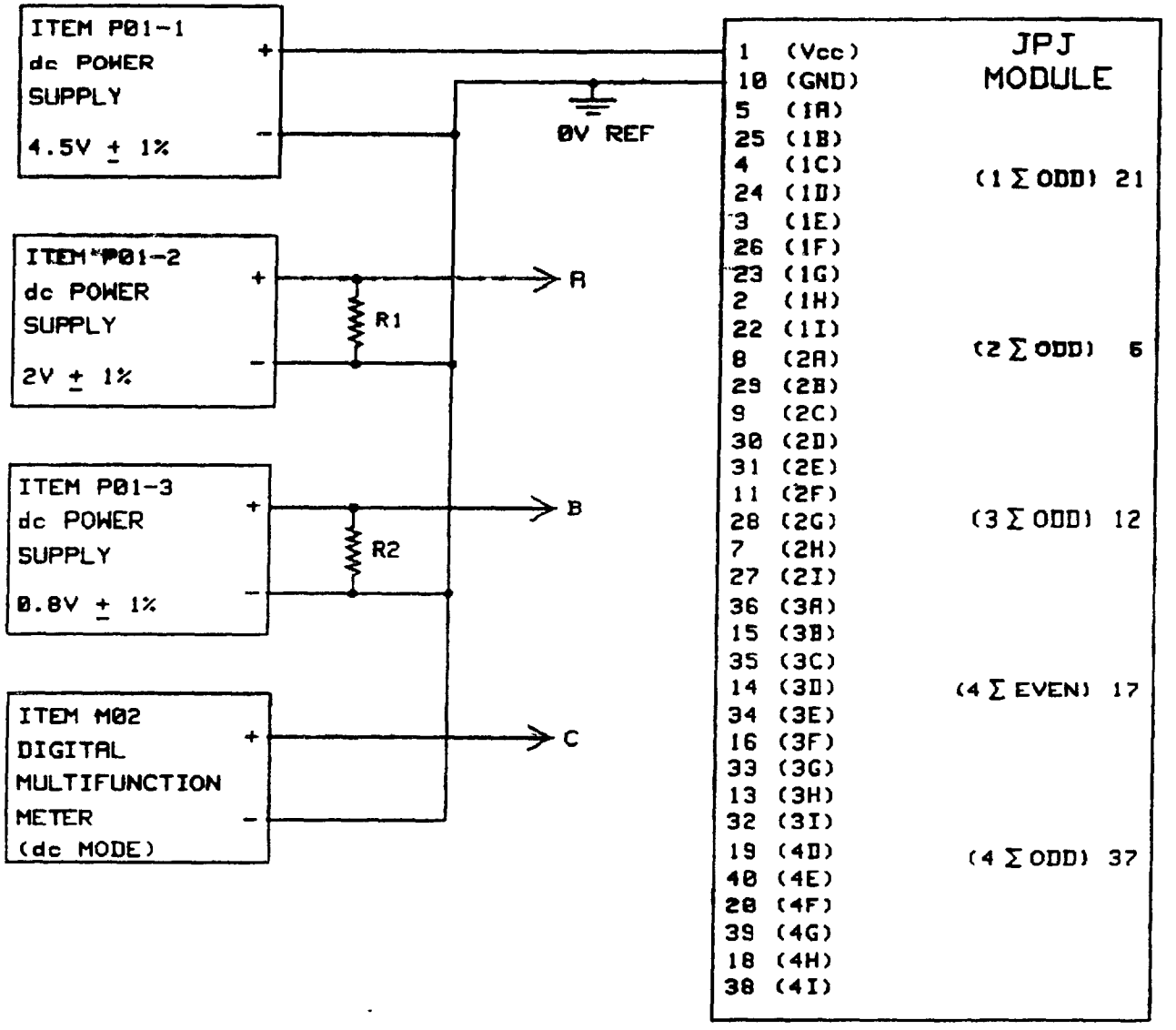
Test cond	A to pin	B to pin	C to pin	1/ Output logic states: 1 sum odd	2 sum odd	3 sum odd	4 sum odd	4 sum even	Test fig
E17	2,3,4,7,11, 14,15,19, 27,28,29, 31,32,35, 36,39,40	5,8,9,13, 16,18,20, 22,23,24, 25,26,30, 33,34,38	2/3/	H	L	H	H	L	11
E18	5,8,14,18, 19,20,23, 24,28,30, 33,36,38	2,3,4,7,9, 11,13,15, 16,22,25, 26,27,29, 31,32,34, 35,39,40		H	H	H	H	L	
E19	3,5,8,14, 15,22,26, 28,30,33, 39,40	2,4,7,9,11, 13,16,18, 19,20,23, 24,25,27, 29,31,32, 34,35,36,38	2/3/	L	H	H	L	H	11

1/ Output logic states:
L = Low level = 0.0 to 0.5V.
H = High level = 2.5 to 4.5V.

2/ For test conditions E1-E19, verification of the output logic states in footnote 1 is obtained by connecting item M02 (point C) to each output pin in succession after input conditions have been accomplished.

3/ 1 sum odd, 2 sum odd, 3 sum odd, 4 sum odd and 4 sum even output logic states are measured at output pins 21, 6, 12, 37 and 17, respectively.

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Check the source to verify that this is the current version before use.

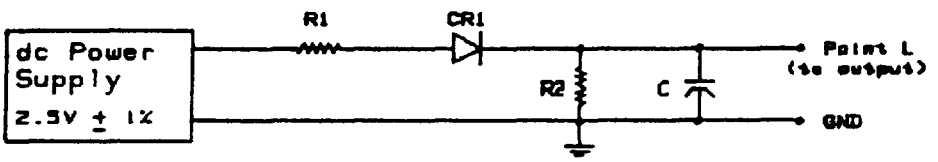


NOTE:

1. R1 = R2 = 51 Ohms ± 5 percent, 1/2W.

FIGURE 11. Functional test circuit for table VII.

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Check the source to verify that this is the currentversion before use.



NOTES:

1. CR1 = 1N4150 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for test type Q.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for test type S.

(c) C not required for test conditions G1-G5 and H1-H5.

(d) The lead length from load circuit to module output pin under test shall be less than 1 inch (25.4mm).

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
21,6,12	6.58k	3.63	0.4
17,37	6.25k	4.0	0.4

1/ All resistors are metal film, ± 1 percent, 1/4 W.

FIGURE 12. Logic load circuit.

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TABLE VIII. Input clamp diode voltage test.

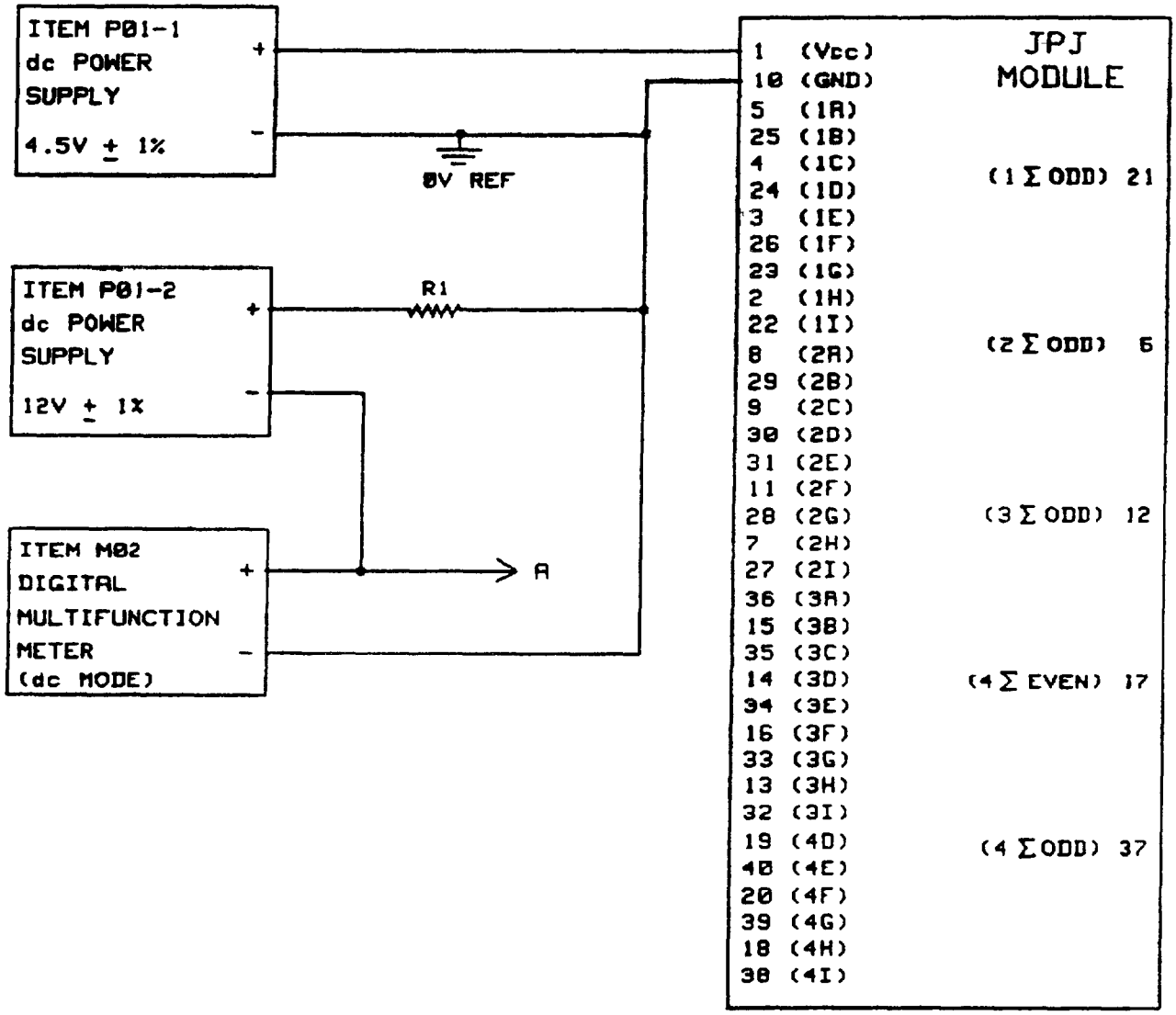
Test cond	Signal name	A to pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Test fig
F1	1A	5	-18.0	-0.4	-1.5	V	13
F2	1B	25					
F3	1C	4					
F4	1D	24					
F5	1E	3					
F6	1F	26					
F7	1G	23					
F8	1H	2					
F9	1I	22					
F10	2A	8					
F11	2B	29					
F12	2C	9					
F13	2D	30					
F14	2E	31					
F15	2F	11					
F16	2G	28					
F17	2H	7					
F18	2I	27					
F19	3A	36					
F20	3B	15	-18.0	-0.4	-1.5	V	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:18Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test - Continued.

Test cond	Signal name	A to pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Test fig
F21	3C	35	-18.0	-0.4	-1.5	V	13
F22	3D	14					
F23	3E	34					
F24	3F	16					
F25	3G	33					
F26	3H	13					
F27	3I	32					
F28	4D	19					
F29	4E	40					
F30	4F	20					
F31	4G	39					
F32	4H	18					
F33	4I	38	-18.0	-0.4	-1.5	V	13

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:18Z
Check the source to verify that this is the current version before use.



NOTE:

1. R1 = 590 Ohms ± 1 percent, 1/4W.

FIGURE 13. Input clamp diode voltage test circuit for table VIII.

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Check the source to verify that this is the current version before use.

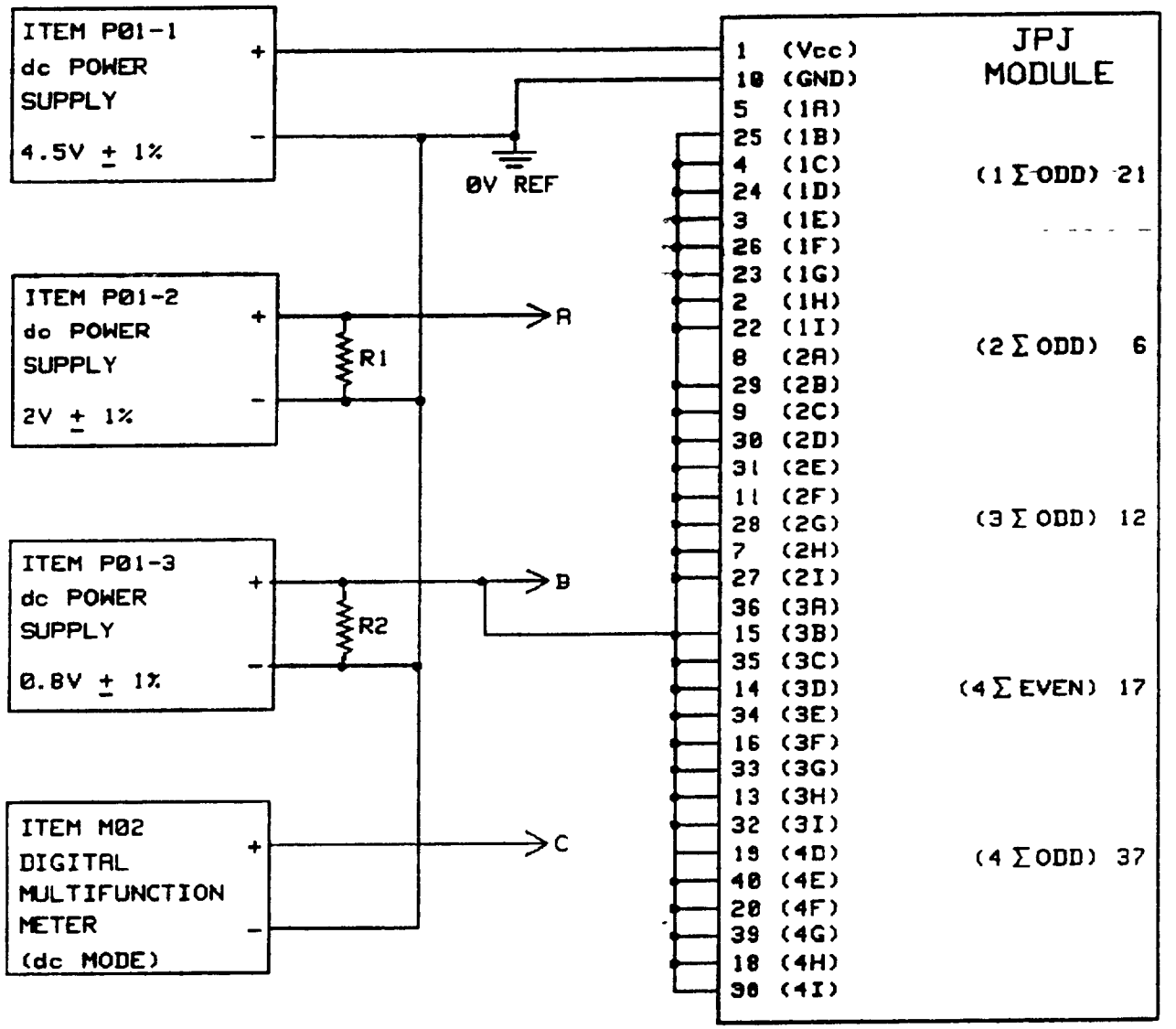
TABLE IX. Output voltage, low level, test.

Test cond	Signal name	A to pin no.	B to pin no.	C to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
G1	1 SUM ODD	OPEN	5,8,36	21	0	400	mV	14
G2	2 SUM ODD			6				
G3	3 SUM ODD			12				
G4	4 SUM ODD	OPEN	5,8,36	37				
G5	4 SUM EVEN	36	5,8	17	0	400	mV	14

TABLE X. Output voltage, high level, test.

Test cond	Signal name	A to pin no.	B to pin no.	C to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
H1	1 SUM ODD	5	8,36	21	2.5	4.5	V	14
H2	2 SUM ODD	8	5,36	6				
H3	3 SUM ODD	36	5,8	12				
H4	4 SUM ODD	36	5,8	37				
H5	4 SUM EVEN	OPEN	5,8,36	17	2.5	4.5	V	14

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Check the source to verify that this is the current version before use.



NOTES:

1. R1 = R2 = 51 Ohms ± 5 percent, 1/2W.
2. Load all outputs with a load as shown on figure 12.

FIGURE 14. Output voltage, low and high level, test circuit for tables IX and X.

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Check the source to verify that this is the current version before use.

TABLE XI. Input current, high level, test.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
J1	1A	2,3,4,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	5	-2	20	uA	15
J2	1B	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	25				
J3	1C	2,3,5,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	4				
J4	1D	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	24				
J5	1E	2,4,5,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	3				
J6	1F	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,27,28,29, 30,31,32,33,34, 35,36,38,39,40	26	-2	20	uA	15

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TABLE XI. Input current, high level, test - Continued.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
J7	1G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	23	-2	20	uA	15
J8	1H	3,4,5,7,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	2				
J9	1I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	22				
J10	2A	2,3,4,5,7,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	8				
J11	2B	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 30,31,32,33,34, 35,36,38,39,40	29				
J12	2C	2,3,4,5,7,8,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	9	-2	20	uA	15

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TABLE XI. Input current, high level, test - Continued.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
J13	2D	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,31,32,33,34, 35,36,38,39,40	30	-2	20	uA	15
J14	2E	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,32,33,34, 35,36,38,39,40	31				
J15	2F	2,3,4,5,7,8,9, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	11				
J16	2G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,29, 30,31,32,33,34, 35,36,38,39,40	28				
J17	2H	2,3,4,5,8,9,11, 13,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	7				
J18	2I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,28,29, 30,31,32,33,34, 35,36,38,39,40	27	-2	20	uA	15

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TABLE XI. Input current, high level, test - Continued.

Test cond	Siganl name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
J19	3A	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,38,39,40	36	-2	20	uA	15
J20	3B	2,3,4,5,7,8,9, 11,13,14,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	15				
J21	3C	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,36,38,39,40	35				
J22	3D	2,3,4,5,7,8,9, 11,13,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	14				
J23	3E	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 35,36,38,39,40	34				
J24	3F	2,3,4,5,7,8,9, 11,13,14,15,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	16	-2	20	uA	15

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:18Z
Check the source to verify that this is the currentversion before use.

TABLE XI. Input current, high level, test - Continued.

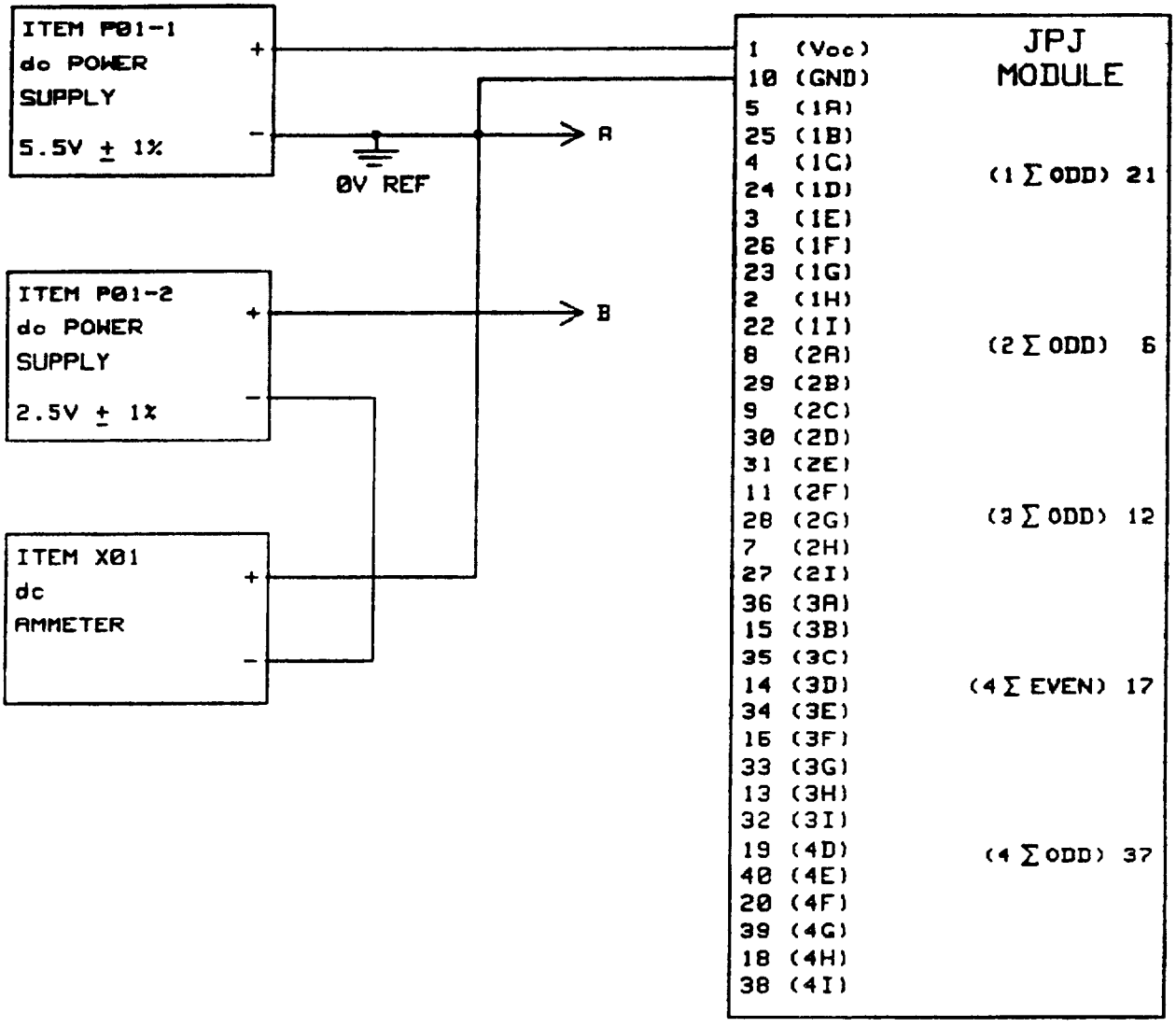
Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
J25	3G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,34, 35,36,38,39,40	33	-2	20	uA	15
J26	3H	2,3,4,5,7,8,9, 11,14,15,16,18, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	13				
J27	3I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,33,34, 35,36,38,39,40	32				
J28	4D	2,3,4,5,7,8,9, 11,13,14,15,16, 18,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	19				
J29	4E	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,36,38,39	40				
J30	4F	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	20	-2	20	uA	15

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TABLE XI. Input current, high level, test - Continued.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
J31	4G	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,36,38,40	39	-2	20	uA	15
J32	4H	2,3,4,5,7,8,9, 11,13,14,15,16, 19,20,22,23,24, 25,26,27,28,29, 30,31,32,33,34, 35,36,38,39,40	18				
J33	4I	2,3,4,5,7,8,9, 11,13,14,15,16, 18,19,20,22,23, 24,25,26,27,28, 29,30,31,32,33, 34,35,36,39,40	38	-2	20	uA	15

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NOTE:

1. Caution: Do not isolate the chassis of item X01 from earth ground.

FIGURE 15. Input current, high level, test circuit for table XI.

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TABLE XII. Input current, low level, test.

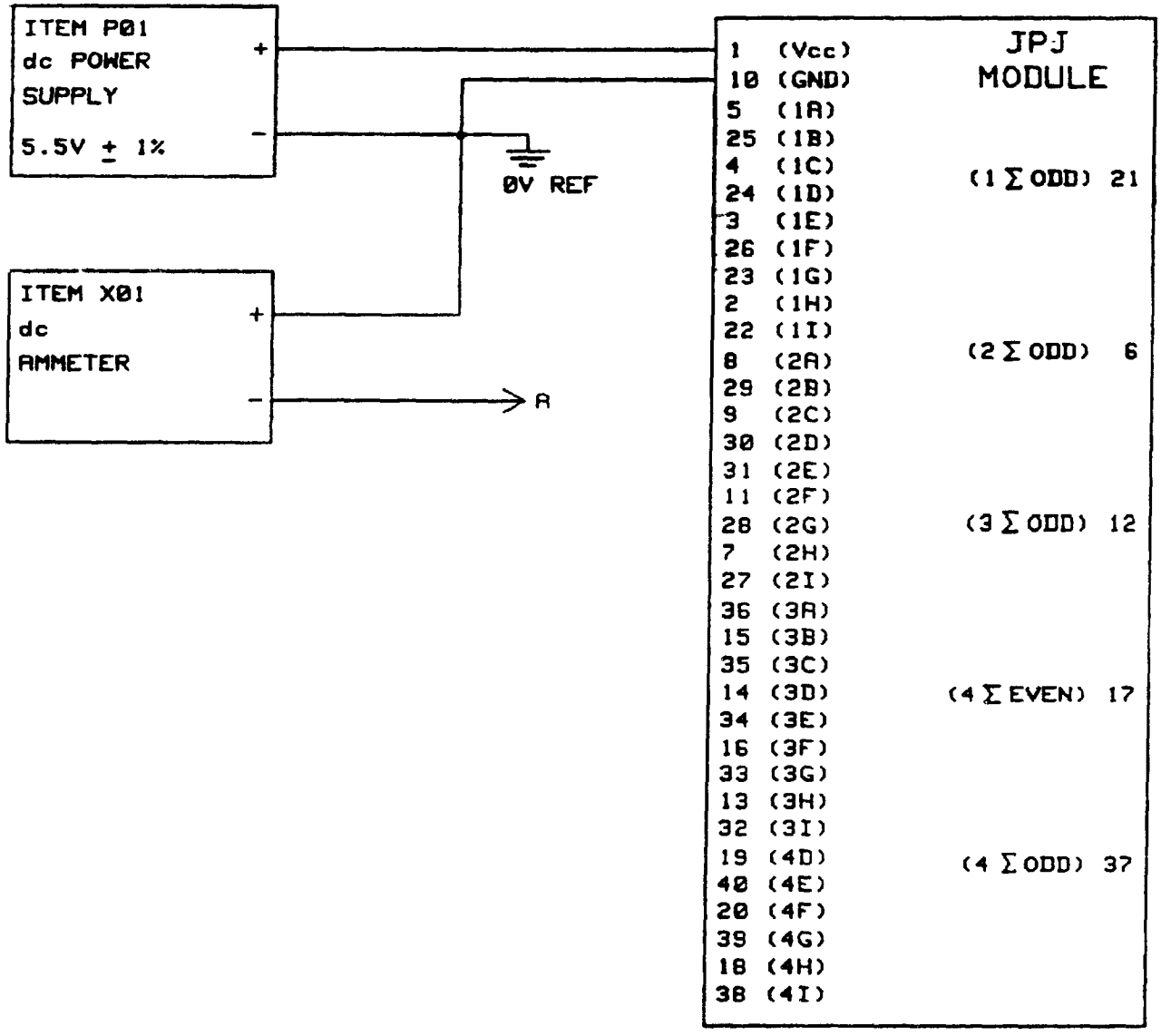
Test cond	Signal name	A to pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
K1	1A	5	0.4	-0.13	-0.37	mA	16
K2	1B	25					
K3	1C	4					
K4	1D	24					
K5	1E	3					
K6	1F	26					
K7	1G	23					
K8	1H	2					
K9	1I	22					
K10	2A	8					
K11	2B	29					
K12	2C	9					
K13	2D	30					
K14	2E	31					
K15	2F	11					
K16	2G	28					
K17	2H	7					
K18	2I	27					
K19	3A	36					
K20	3B	15	0.4	-0.13	-0.37	mA	16

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TABLE XII. Input current, low level, test - Continued.

Test cond	Signal name	A to pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Test fig
K21	3C	35	0.4	-0.13	-0.37	mA	16
K22	3D	14					
K23	3E	34					
K24	3F	16					
K25	3G	33					
K26	3H	13					
K27	3I	32					
K28	4D	19					
K29	4E	40					
K30	4F	20					
K31	4G	39					
K32	4H	18					
K33	4I	38	0.4	-0.13	-0.37	mA	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:18Z
Check the source to verify that this is the current version before use.



NOTE:

1. CAUTION: Do not isolate the chassis of item X01 from earth ground.

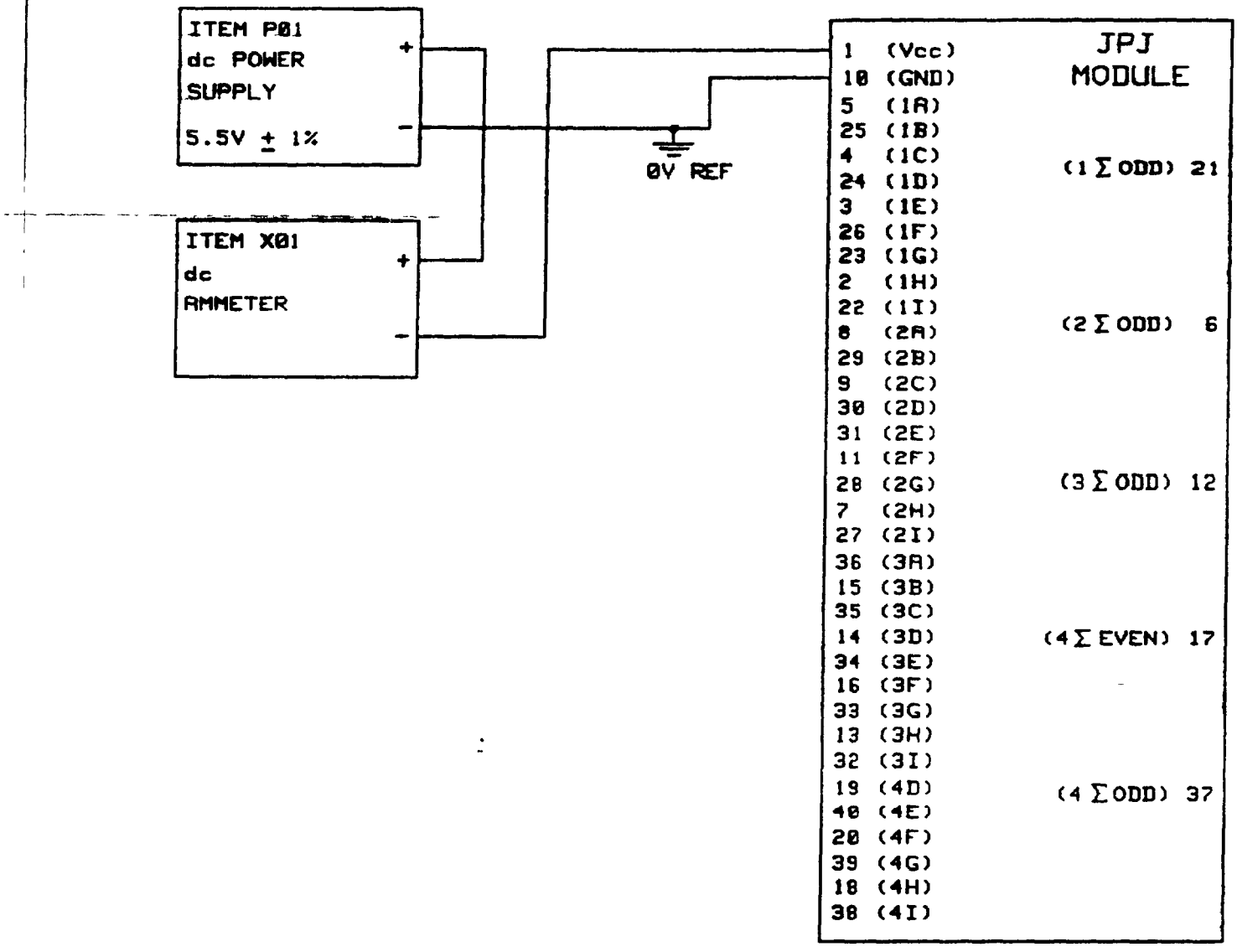
FIGURE 16. Input current, low level, test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Test fig
		Min	Max		
L	Power supply current	0	108	mA	17

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NOTE:

1. Caution: Do not isolate the chassis of item X01 from earth ground.

FIGURE 17. Power supply current test circuit for table XIII.

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TABLE XIV. Propagation delay time, tPLH test.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C Min	Initial limits 25 ± 5°C Max	Unit	Test fig
M1	1A to 1 sum odd	tPLH CL=50pF	5	21	2,22,23	2	55	ns	18, 19
M2	1B to 1 sum odd		25	21	3,4,5,24, 26				
M3	1C to 1 sum odd		4	21	3,5,24,25, 26				
M4	1C to 1 sum odd		4	21	2,3,22,23, 24,25,26				
M5	1D to 1 sum odd		24	21	4,5,25				
M6	2A to 2 sum odd		8	6	7,27,28				
M7	2B to 2 sum odd		29	6	8,9,11,30, 31				
M8	2C to 2 sum odd		9	6	8,11,29, 30,31				
M9	2C to 2 sum odd		9	6	7,11,27, 28,29,30, 31				
M10	2D to 2 sum odd	tPLH CL=50pF	30	6	8,9,29	2	55	ns	18, 19

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TABLE XIV. Propagation delay time, tPLH test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
M11	3A to 3 sum odd	tPLH CL=50pF	36	12	13,32,33	2	55	nS	18, 19
M12	3B to 3 sum odd		15	12	14,16,34, 35,36				
M13	3C to 3 sum odd		35	12	14,15,16, 34,36				
M14	3C to 3 sum odd		35	12	13,14,15, 16,32,33, 34				
M15	3D to 3 sum odd		14	12	15,35,36				
M16	4D to 4 sum odd		19	37	5,8,18,36, 38,39,40				
M17	4E to 4 sum odd		40	37	18,19,20, 38,39				
M18	4F to 4 sum odd		20	37	18,38,39				
M19	4D to 4 sum even	tPLH CL=50pF	19	17	5,8,18,20, 36,38,39, 40	2	55	nS	18, 19

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TABLE XIV. Propagation delay time, tPLH test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
M20	4E to 4 sum even	tPLH CL=50pF	40	17	5,8,18,19, 20,36,38, 39	2	55	nS	18, 19
M21	4F to 4 sum even		20	17	19,40	2	55	nS	18, 19
M22	1D to 4 sum odd		24	37	2,3,8,19, 20,22,23, 26,36,40	2	110	nS	18, 19
M23	2D to 4 sum odd		30	37	5,7,11,19, 20,27,28, 31,36,40				
M24	3D to 4 sum odd		14	37	5,8,13,16, 19,20,32, 33,34,40				
M25	1G to 4 sum even		23	17	2,3,8,18, 19,20,22, 24,26,36, 38,39,40				
M26	2G to 4 sum even		28	17	5,7,11,18, 19,20,27, 30,31,36, 38,39,40				
M27	3G to 4 sum even	tPLH CL=50pF	33	17	13,14,16, 18,19,20, 32,34,38, 39,40	2	110	nS	18, 19

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TABLE XIV. Propagation delay time, tPLH test - Continued.

Test no	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
M28	1A to 1 sum odd	tPLH CL=150pF	5	21	2,22,23	2	65	nS	18, 19
M29	1B to 1 sum odd		25	21	3,4,5,24, 26				
M30	1C to 1 sum odd		4	21	3,5,24,25, 26				
M31	1C to 1 sum odd		4	21	2,3,22,23, 24,25,26				
M32	1D to 1 sum odd		24	21	4,5,25				
M33	2A to 2 sum odd		8	6	7,27,28				
M34	2B to 2 sum odd		29	6	8,9,11,30, 31				
M35	2C to 2 sum odd		9	6	8,11,29, 30,31				
M36	2C to 2 sum odd	tPLH CL=150pF	9	6	7,11,27, 28,29,30, 31	2	65	nS	18, 19

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TABLE XIV. Propagation delay time, tPLH test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
M37	2D to 2 sum odd	tPLH CL=150pF	30	6	8,9,29	2	65	nS	18, 19
M38	3A to 3 sum odd		36	12	13,32,33				
M39	3B to 3 sum odd		15	12	14,16,34, 35,36				
M40	3C to 3 sum odd		35	12	14,15,16, 34,36				
M41	3C to 3 sum odd		35	12	13,14,15, 16,32,33, 34				
M42	3D to 3 sum odd		14	12	15,35,36				
M43	4D to 4 sum odd		19	37	5,8,18,36, 38,39,40				
M44	4E to 4 sum odd		40	37	18,19,20, 38,39				
M45	4F to 4 sum odd	tPLH CL=150pF	20	37	18,38,39	2	65	nS	18, 19

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TABLE XIV. Propagation delay time, tPLH test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
M46	4D to 4 sum even	tPLH CL=150pF	19	17	5,8,18,20, 36,38,39, 40	2	65	nS	18, 19
M47	4E to 4 sum even		40	17	5,8,18,19, 20,36,38, 39				
M48	4F to 4 sum even		20	17	19,40	2	65	nS	18, 19
M49	1D to 4 sum odd		24	37	2,3,8,19, 20,22,23, 26,36,40	2	130	nS	18, 19
M50	2D to 4 sum odd		30	37	5,7,11,19, 20,27,28, 31,36,40				
M51	3D to 4 sum odd		14	37	5,8,13,16, 19,20,32, 33,34,40				
M52	1G to 4 sum even		23	17	2,3,8,18, 19,20,22, 24,26,36, 38,39,40				
M53	2G to 4 sum even		28	17	5,7,11,18, 19,20,27, 30,31,36, 38,39,40				
M54	3G to 4 sum even	tPLH CL=150pF	33	17	13,14,16, 18,19,20, 32,34,38, 39,40	2	130	nS	18, 19

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TABLE XV. Propagation delay time, tPHL test.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
N1	1E to 1 sum odd	tPHL CL=50pF	3	21	4,5,24,25, 26	2	50	nS	18, 19
N2	1F to 1 sum odd		26		3,4,5,24, 25				
N3	1G to 1 sum odd		23		2,3,4,5, 24,25,26				
N4	1H to 1 sum odd		2		3,22,23, 24,26				
N5	1I to 1 sum odd		22	21	3,24,26				
N6	2E to 2 sum odd		31	6	8,9,11, 29,30				
N7	2F to 2 sum odd		11		8,9,29, 30,31				
N8	2G to 2 sum odd		28		7,8,9, 11,29,30, 31				
N9	2H to 2 sum odd	tPHL CL=50pF	7	6	11,27,28, 30,31	2	50	nS	18, 19

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TABLE XV. Propagation delay time, tPHL test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
N10	2I to 2 sum odd	tPHL CL=50pF	27	6	11,30,31	2	50	nS	18, 19
N11	3E to 3 sum odd		34	12	14,15,16, 35,36				
N12	3F to 3 sum odd		16	12	14,15,34, 35,36				
N13	3G to 3 sum odd		33	12	13,14,15, 16,34,35, 36				
N14	3H to 3 sum odd		13	12	14,16,32, 33,34				
N15	3I to 3 sum odd		32	12	14,16,34				
N16	4D to 4 sum odd		19	37	5,8,18,36, 38,39,40				
N17	4G to 4 sum odd		39	37	5,8,36				
N18	4H to 4 sum odd	tPHL CL=50pF	18	37	5,8,36,38, 39	2	50	nS	18, 19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:18Z
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TABLE XV. Propagation delay time, tPHL test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
N19	4I to 4 sum odd	tPHL CL=50pF	38	37	5,8,18,36, 39	2	50	nS	18, 19
N20	4D to 4 sum odd		19	17	5,8,20,36				
N21	4G to 4 sum even		39	17	5,8,18,19, 20,36,38, 40				
N22	4H to 4 sum even		18	17	5,8,19,20, 36,38,39, 40				
N23	4I to 4 sum even		38	17	5,8,18,19, 20,36,39, 40	2	50	nS	18, 19
N24	1D to 4 sum odd		24	37	2,3,8,19, 20,22,23, 26,36,40	2	100	nS	18, 19
N25	2D to 4 sum odd		30	37	5,7,11,19, 20,27,28, 31,36,40				
N26	3D to 4 sum odd		14	37	5,8,13,16, 19,20,32, 33,34,40				
N27	1G to 4 sum even	tPHL CL=50pF	23	17	2,3,8,18, 19,20,22, 24,26,36, 38,39,40	2	100	nS	18, 19

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TABLE XV. Propagation delay time, tPHL test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
N28	2G to 4 sum even	tPHL CL=50pF	28	17	2,3,8,18, 19,20,22, 24,26,36, 38,39,40	2	100	nS	18, 19
N29	3G to 4 sum even	tPHL CL=50pF	33	17	13,14,16, 18,19,20, 32,34,38, 39,40	2	100	nS	18, 19
N30	1E to 1 sum odd	tPHL CL=150pF	3	21	4,5,24,25, 26	2	60	nS	18, 19
N31	1F to 1 sum odd		26	21	3,4,5,24, 25				
N32	1G to 1 sum odd		23	21	2,3,4,5, 24,25,26				
N33	1H to 1 sum odd		2	21	3,22,23, 24,26				
N34	1I to 1 sum odd		22	21	3,24,26				
N35	2E to 2 sum odd	tPHL Cl=150pF	31	6	8,9,11,29, 30	2	60	nS	18, 19

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TABLE XV. Propagation delay time, tPHL test - Continued.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
N36	2F to 2 sum odd	tPHL CL=150pF	11	6	8,9,29,30, 31	2	60	nS	18, 19
N37	2G to 2 sum odd		28	6	7,8,9,11, 29,30,31				
N38	2H to 2 sum odd		7	6	11,27,28, 30,31				
N39	2I to 2 sum odd		27	6	11,30,31				
N40	3E to 3 sum odd		34	12	14,15,16, 35,36				
N41	3F to 3 sum odd		16	12	14,15,34, 35,36				
N42	3G to 3 sum odd		33	12	13,14,15, 16,34,35, 36				
N43	3H to 3 sum odd		13	12	14,16,32, 33,34				
N44	3I to 3 sum odd	tPHL CL=150pF	32	12	14,16,34	2	60	nS	18, 19

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TABLE XV. Propagation delay time, tPHL test - Continued.

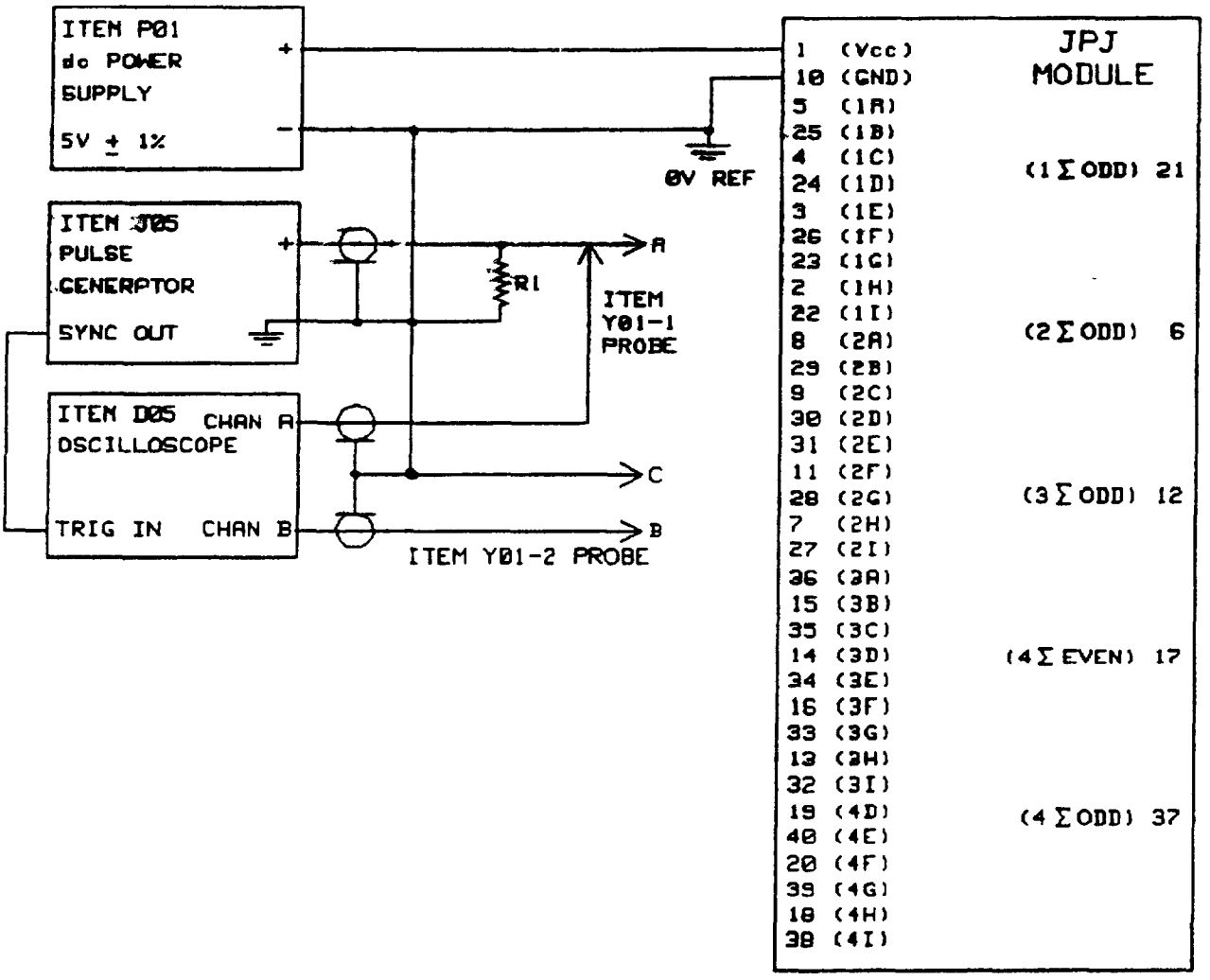
Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
N45	4D to 3 sum odd	tPHL CL=150pF	19	37	5,8,18,36, 38,39,40	2	60	nS	18, 19
N46	4G to 4 sum odd		39	37	5,8,36				
N47	4H to 4 sum odd		18	37	5,8,36,38, 39				
N48	4I to 4 sum odd		38	37	5,8,18,36, 39				
N49	4D to 4 sum even		19	17	5,8,20,36				
N50	4G to 4 sum even		39	17	5,8,18,19, 20,36,38, 40				
N51	4H to 4 sum even		18	37	5,8,19,20, 36,38,39, 40				
N52	4I to 4 sum even		38	17	5,8,18,19, 20,36,39, 40	2	60	nS	18, 19
N53	1D to 4 sum odd	tPHL CL=150pF	24	37	2,3,8,19, 20,22,23, 26,36,40	2	120	nS	18, 19

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TABLE XV. Propagation delay time, tPHL test - Continued.

Test Cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
N54	2D to 4 sum odd	tPHL CL=150pF	30	37	5,7,11,19, 20,27,28, 31,36,40	2	120	nS	18, 19
N55	3D to 4 sum odd		14	37	5,8,13,16, 19,20,32, 33,34,40				
N56	1G to 4 sum even		23	17	2,3,8,18, 19,20,22, 24,26,36, 38,39,40				
N57	2G to 4 sum even		28	17	5,7,11,18, 19,20,27, 30,31,38, 36,39,40				
N58	3G to 4 sum even	tPHL CL=150pF	33	17	13,14,16, 18,19,20, 32,34,38, 39,40	2	120	nS	18, 19

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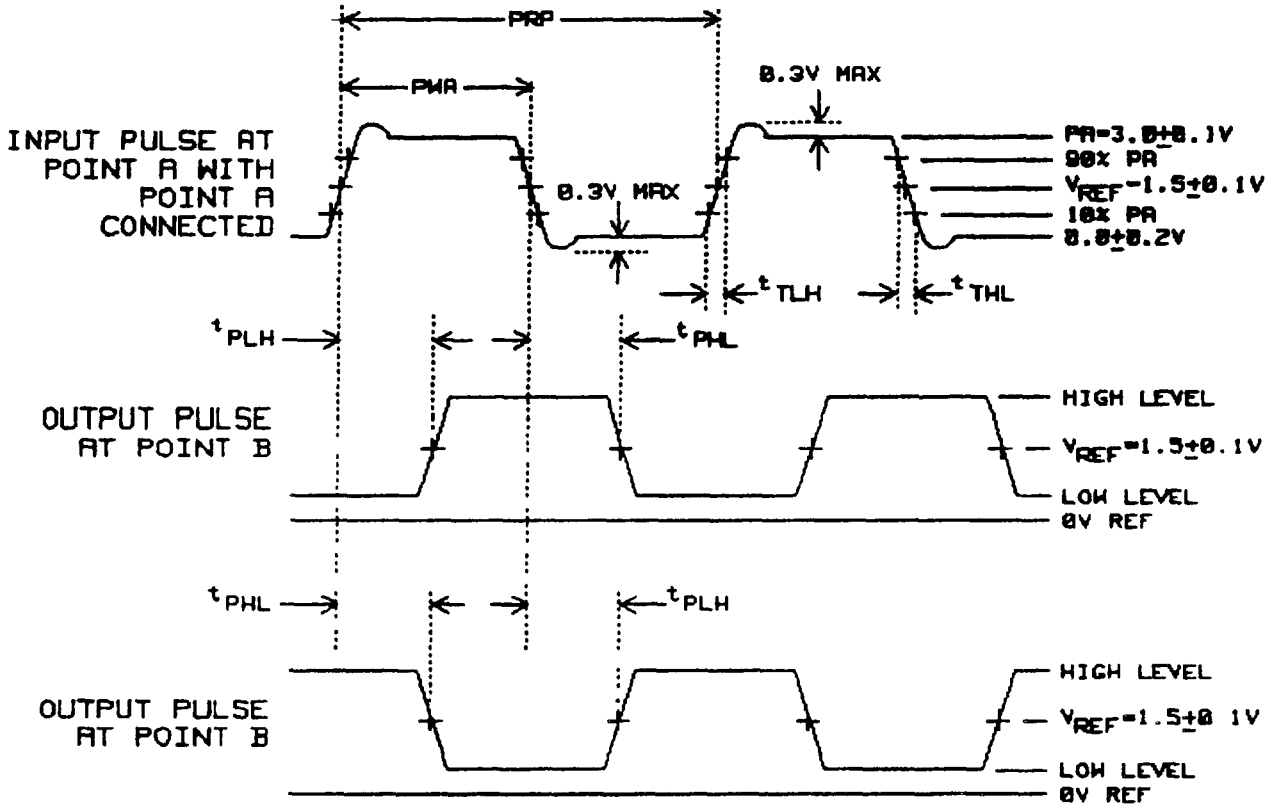


NOTES:

1. R1 = 51.1 Ohms ± 1 percent, 1/4W.
2. Load all outputs with a load as shown on figure 12.

FIGURE 18. Propagation delay time (tPLH and tPHL) test circuit for tables XIV and XV.

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NOTES:

1. Input pulse parameters:
   tTLH = tTHL = 6 ± 2 ns.
   PWA = 500 ± 50 ns.
   PRP = 1000 ± 100 ns.

2. Before making the measurements, verify that the output pulse high level is ≥ 2.5V and the low level is ≤ 0.5V.

3. The lead lengths shall be less than 1 inch (25.4 mm).

4. All voltages are referenced to ground.

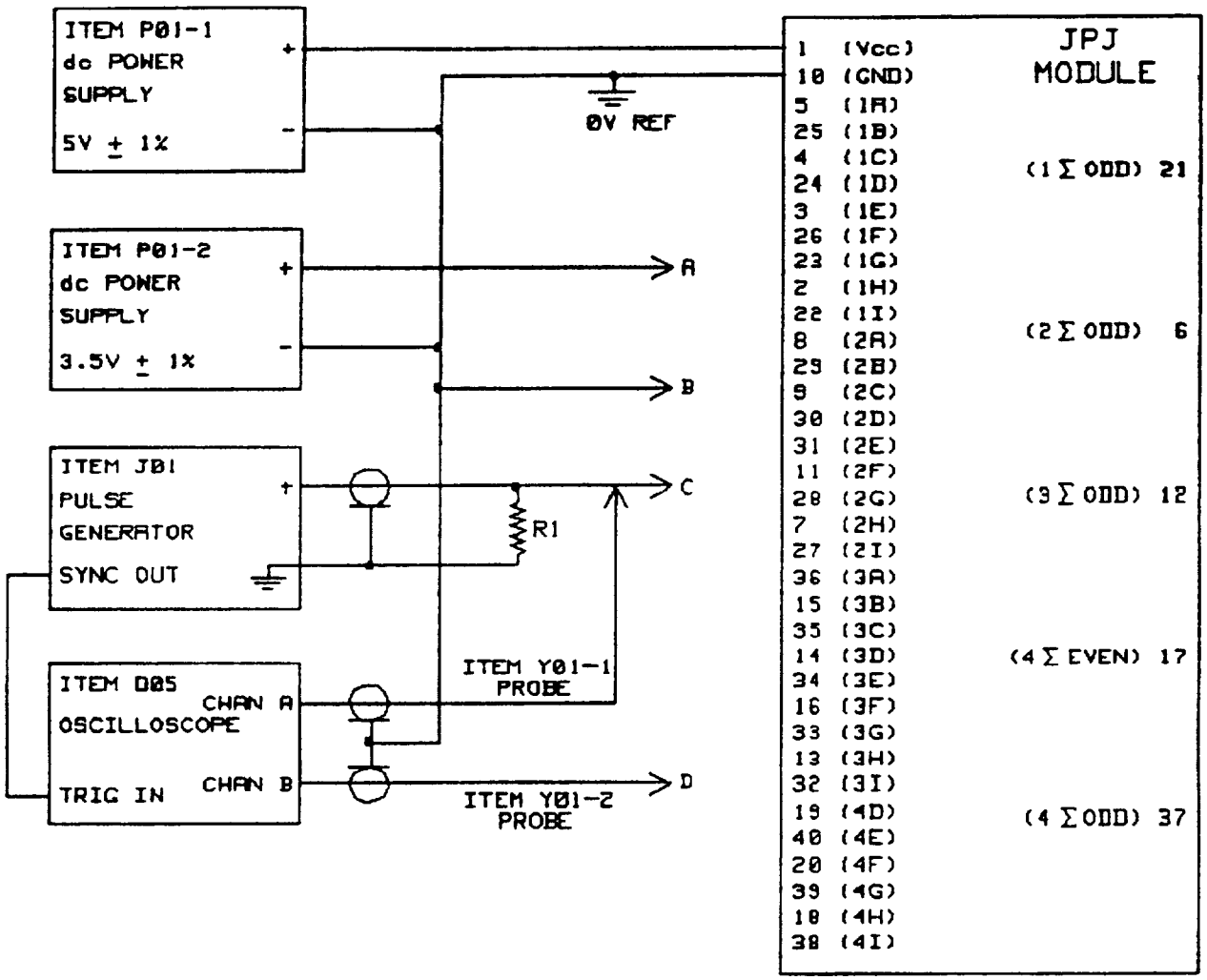
FIGURE 19. Propagation delay time (tPLH and tPHL) test circuit waveforms for tables XIV and XV.

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TABLE XVI. Output crosstalk (low level and high level) test.

Test and	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1 sum add output crosstalk, low level	2,3,4,7,9,11,13, 14,15,16,18,20,22, 23,24,25,26,27,28, 29,30,31,32,33,34, 35,36,39,40	5	8,19,36	21	0	500	mV	20, 21
P2	2 sum add output crosstalk, low level		8	5,19,36	6	0	500	mV	20, 21
P3	3 sum add output crosstalk, low level	2,3,4,7,9,11,13, 14,15,16,18,20,22, 23,24,25,26,27,28, 29,30,31,32,33,34, 35,36,39,40	36	5,8,19	12	0	500	mV	20, 21
Q1	1 sum add output crosstalk, high level	3,7,9,11,13,14,15, 16,18,20,27,28,29, 30,31,32,33,34,35, 36,39,40,	2,3,4,22,23, 24,25,26	8,19,36	21	0	400	mV	20, 21
Q2	2 sum add output crosstalk, high level	2,3,4,8,13,14,15, 16,18,20,22,23,24, 25,26,32,33,34,35, 36,39,40	7,9,11,27,28, 29,30,31	5,19,36	6	0	400	mV	20, 21
Q3	3 sum add output crosstalk,	2,3,4,7,9,11,18, 20,22,23,24,25,26, 27,28,29,30,31,36, 38,39,40	13,14,15,16, 32,33,34,35	5,8,19	12	0	400	mV	20, 21

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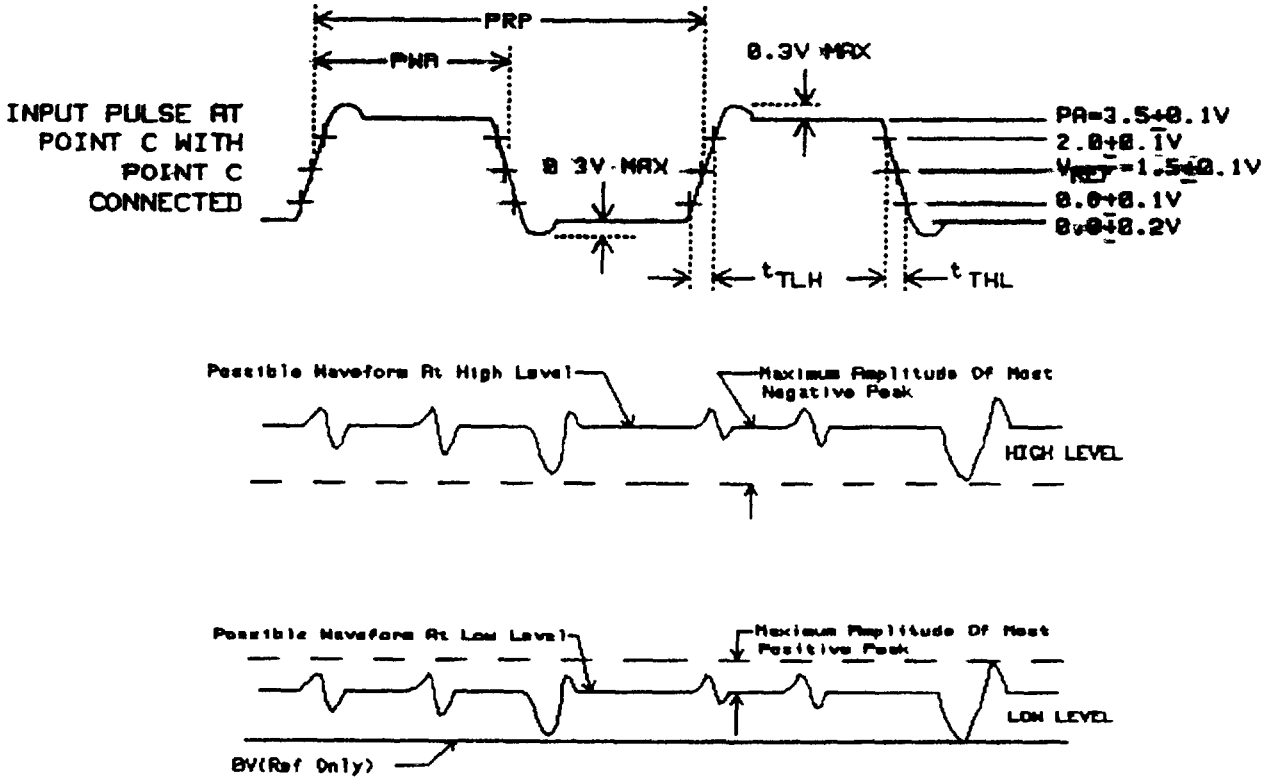


NOTES:

1. R1 = 51.1 Ohms ± 1 percent, 1/4W.
2. Load all outputs with a load as shown on figure 13 with the exception that CL = 50 ± 5pF.
3. The test circuit waveforms shall be as indicated on figure 21.

FIGURE 20. Output crosstalk, low and high level, test circuit for table XVI.

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NOTES:

1. Input pulse parameters:
   PRP = 1000 ± 100 ns.
   PWA = 500 ± 50 ns.
   tTLH = tTHL = 6 ± 2 ns (verified at each input pin)

2. The test circuit shall be as indicated on figure 20.

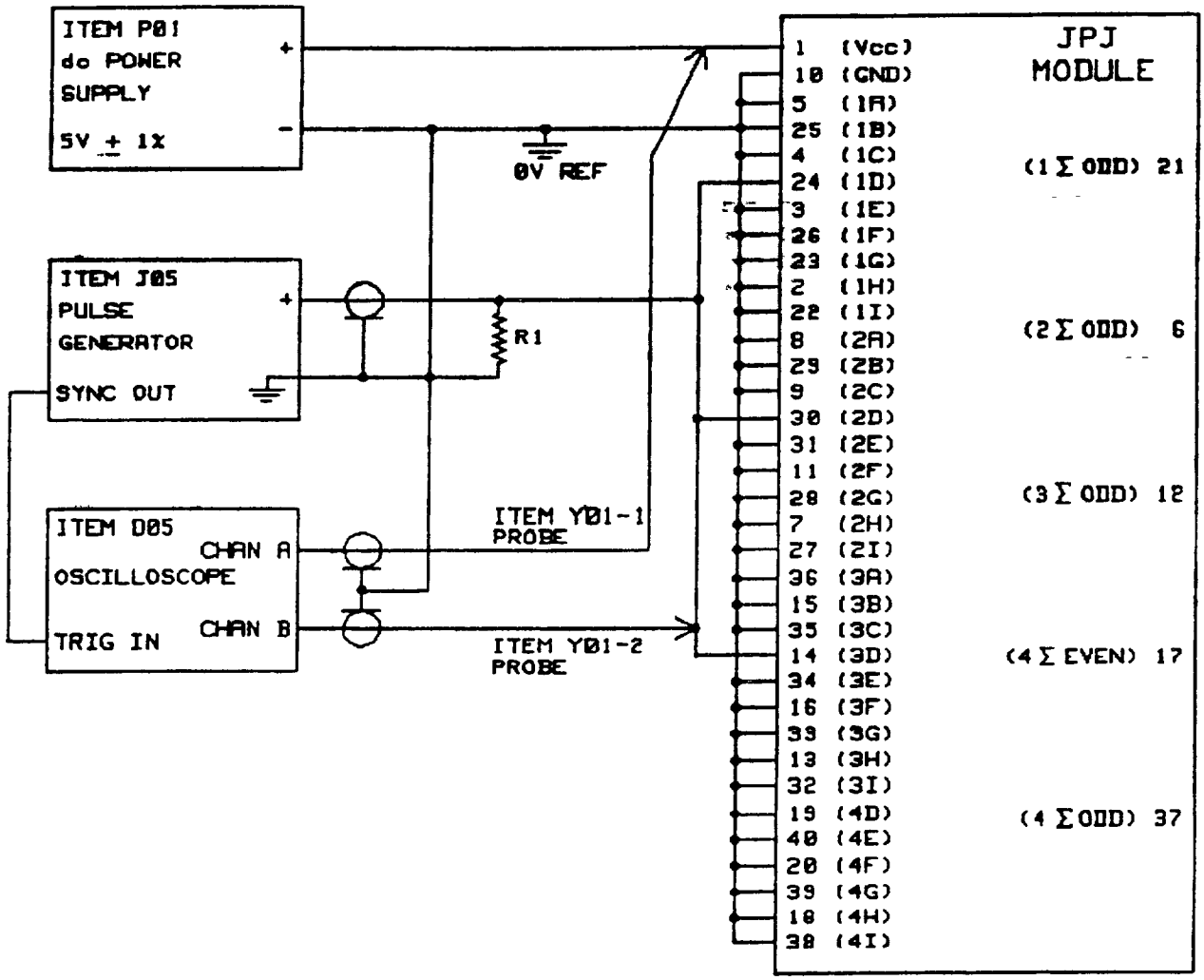
FIGURE 21. Output crosstalk test circuit waveforms (low and high level) for table XVI.

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TABLE XVII. Power supply transient amplitude test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Figs
		Min	Max		
R	PSTA	0	400	mV	22,23

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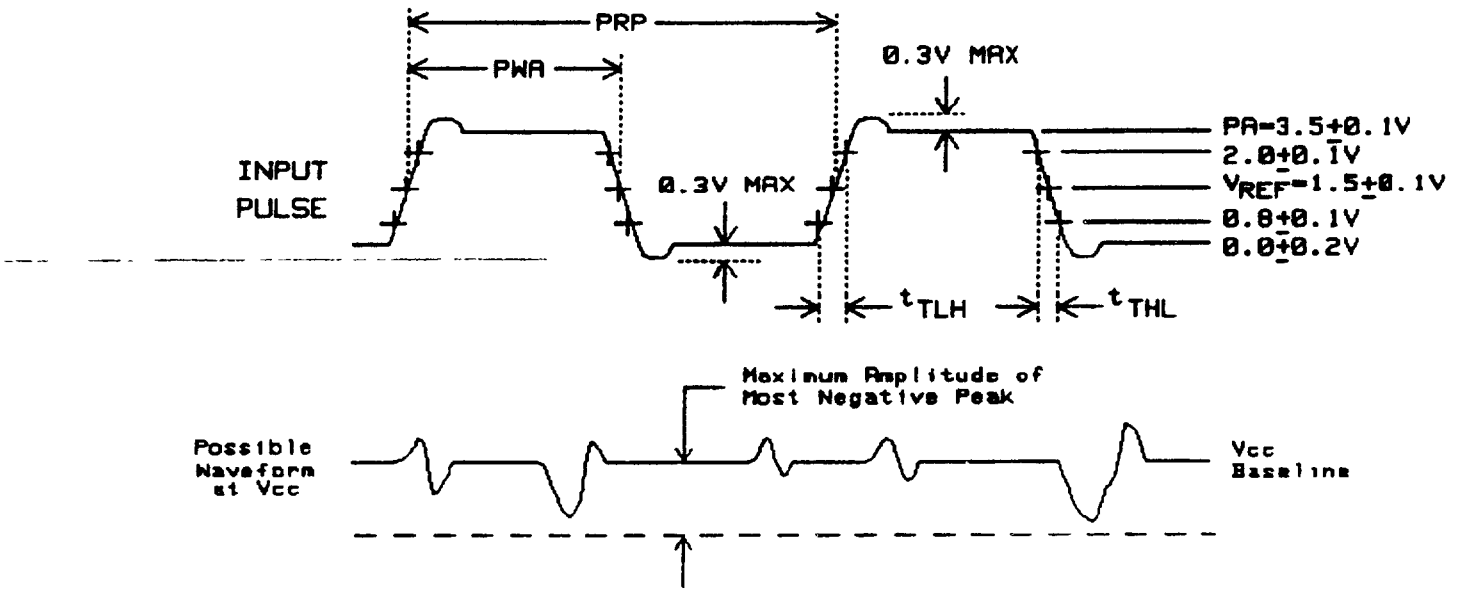


NOTE:

1. R1 = 51.1 Ohms ± 1 percent, 1/4W.

FIGURE 22. Power supply transient amplitude test circuit for table XVII.

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NOTE:

1. Input pulse parameters:
   PRP = 1000 ± 100 ns.
   PWA = 500 ± 50 ns.
   tTLH = tTHL = 6 ± 2 ns (verified at each input pin).

FIGURE 23. Power supply transient amplitude waveform for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Signal name	A to pin no.	B to pin no.	25 ± 5°C Initial limits Min	Max	Units	Test fig
S	PSFC	OPEN	1	1.6	3.0	uF	24

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TABLE XIX. Input capacitance test.

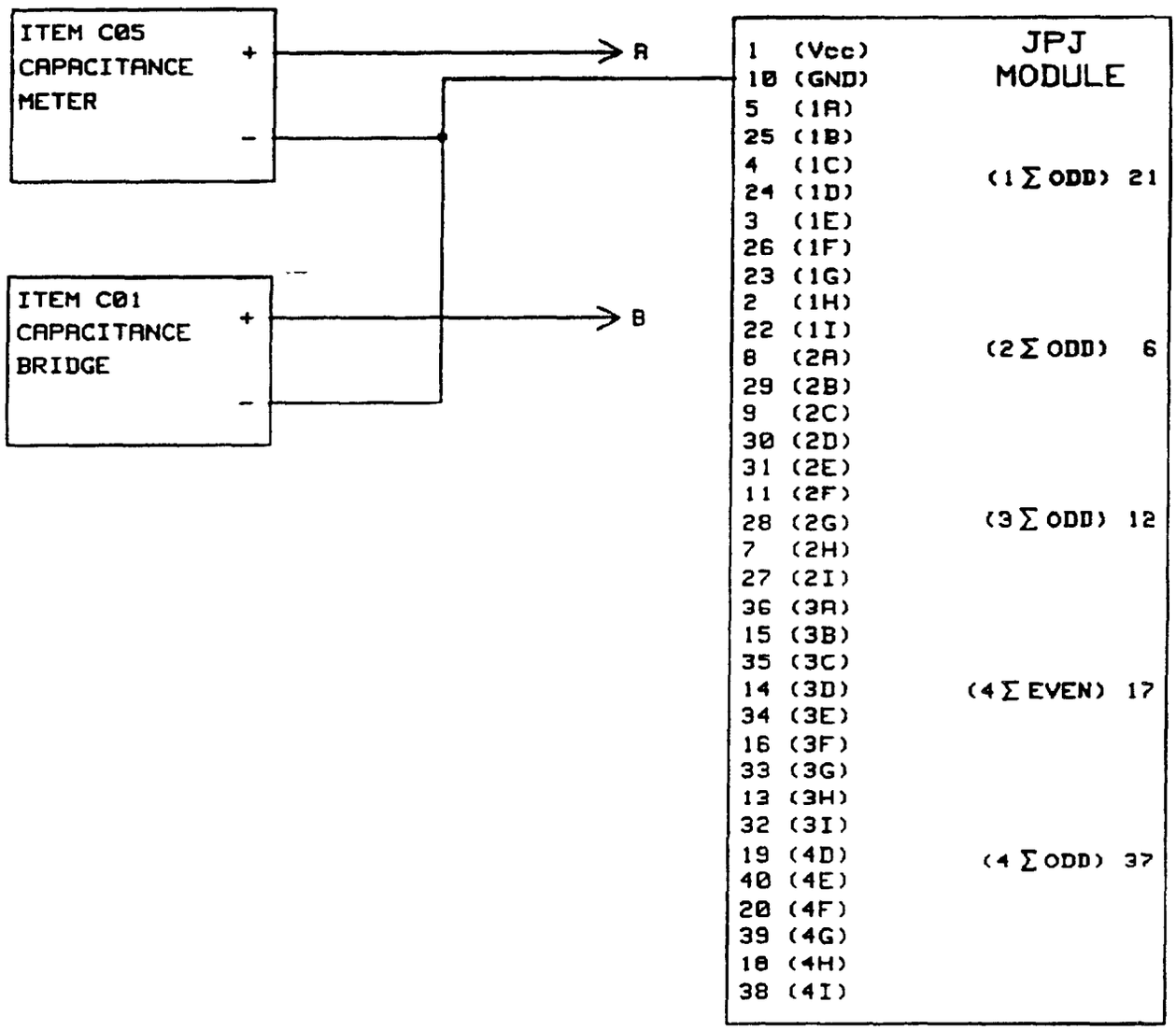
Test cond	Signal name	A to pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Test fig
T1	1A	5	2	15	pF	24
T2	1B	25				
T3	1C	4				
T4	1D	24				
T5	1E	3				
T6	1F	26				
T7	1G	23				
T8	1H	2				
T9	1I	22				
T10	2A	8				
T11	2B	29				
T12	2C	9				
T13	2D	30				
T14	2E	31				
T15	2F	11				
T16	2G	28				
T17	2H	7				
T18	2I	27				
T19	3A	36				
T20	3B	15	2	15	pF	24

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TABLE XIX. Input capacitance test - Continued.

Test cond	Signal name	A to pin no.	25 ± 5°C Initial limits Min	Max	Units	Test fig
T21	3C	35	2	15	pF	24
T22	3D	14				
T23	3E	34				
T24	3F	16				
T25	3G	33				
T26	3H	13				
T27	3I	32				
T28	4D	19				
T29	4E	40				
T30	4F	20				
T31	4G	39				
T32	4H	18				
T33	4I	38	2	15	pF	24

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   Vpk = 0.3 V maximum.

FIGURE 24. Power supply filter and input capacitance test circuit for tables XVIII and XIX.

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